

The Role of Text, Annotations, and Visual Encodings in Shaping Confirmation Bias
During Data Visualization Interpretation

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Abstract

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People rely on data for analysis and decision-making, and data visualizations are essential tools that enhance the efficiency of these processes. However, the process of interpreting data visualizations and making decisions is susceptible to confirmation bias – a mechanism that leads to different interpretations of information when it is consistent or inconsistent with one’s prior beliefs. In the context of data visualization, this bias leads individuals to overemphasize visual patterns that align with their existing beliefs while devaluing data that contradicts them, potentially resulting in skewed analyses and suboptimal decision-making outcomes. The goal of this thesis is to understand how confirmation bias manifests in data visualization interpretation and to design techniques that encourage individuals to also pay attention to patterns that contradict their prior beliefs. Towards this goal, this thesis will (1) examine how individuals mentally represent their prior beliefs using visualization, a critical step to effectively measuring confirmation bias; (2) investigate how the design of *static* data visualizations such as textual summaries and visual annotations can asymmetrically influence confirmation bias; (3) explore how confirmation bias manifest through interactions with data visualization, and (4) design and evaluate interventions that leverage visual encoding manipulations to mitigate bias and promote more balanced, informed reasoning with data visualizations.

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Chapter 1

Introduction

People can perceive many patterns from a single data visualization. For instance, Figure 1.1 shows mortality rates per 100,000 people according to COVID-19 vaccination level across 5 days. Individuals who support vaccination might highlight May 22nd and May 24th, as shown in Figure 1.1 (a), which exhibit an overall negative correlation between fuller vaccination rates and mortality rates. Conversely, those against vaccination might focus on May 23rd and May 25th, as depicted in Figure 1.1 (b), where there is a positive correlation between fuller vaccination rates and mortality rates. This preferred interpretation for an individual is likely influenced by confirmation bias, which is an ubiquitous tendency of searching for, and interpreting information that aligns with prior beliefs [88]. In data visualization, confirmation bias may cause individuals to overemphasize visual patterns that support their prior beliefs while overlooking or disregarding those that contradict them, thereby hindering objective interpretations of the data [140, 6].

These varying interpretations highlight the potential influence of human prior beliefs as well as suggest the potential of visualization design choices to affect the manifestation of confirmation bias. From the human perspective, individuals with strong prior beliefs are more likely to exhibit confirmation bias [15]. From the visu-

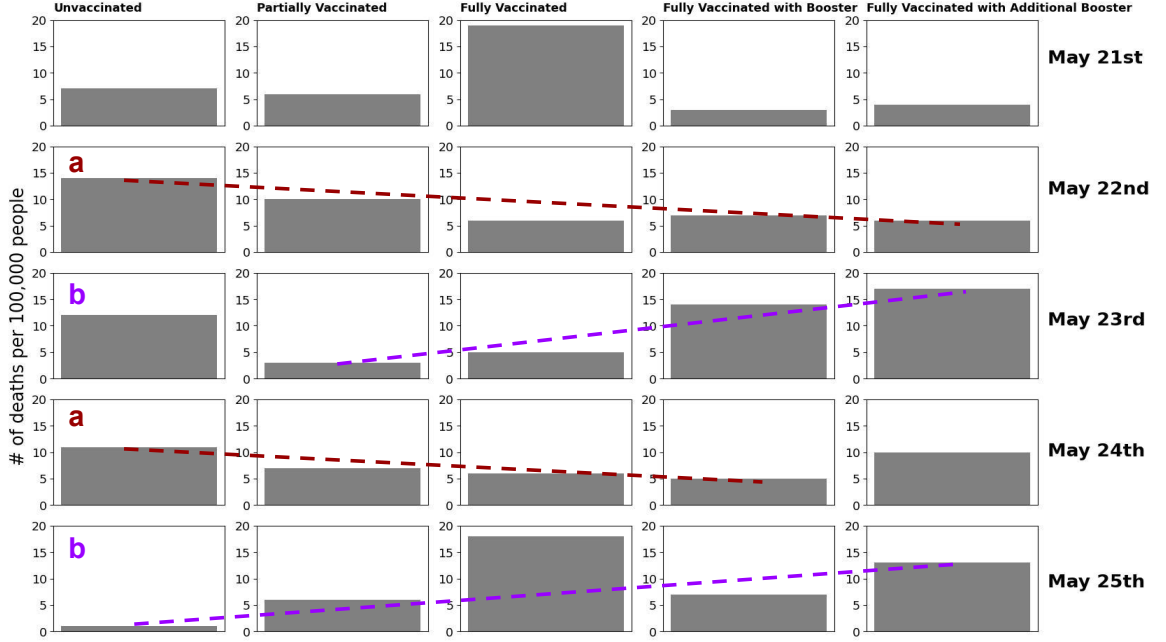


Figure 1.1: An ambiguous data visualization displaying an equal number of patterns showing both positive and negative correlations between fuller vaccination status and mortality rate.

alization design perspective, design choices such as the inclusion of *visual encodings*, *annotations*, and *textual elements* can shape the patterns they perceive in a visualization. For instance, visual *encodings* such as color, texture, and geometry change the accuracy with which people are able to decode information from a visualization [26]. Building on these foundational encodings, *annotations* – such as bounding boxes or trend lines – can further emphasize specific visual patterns. When these emphasized patterns align with a viewer’s prior beliefs, they may reinforce or even amplify biased interpretations [6]. Additionally, recent work in visualization research has increasingly recognized *text* as a critical component of visualization design that might impact the data interpretation [110, 111]. Textual elements in visualizations such as the presence and framing of titles and captions can significantly impact readers’ information recall, reasoning, and decision-making processes. Prior research shows that viewers are more likely to recall information conveyed by biased titles than the actual content of a visualization [66, 65]. Captions that highlight particular visual features can lead viewers

to treat those features as the main takeaway, while mismatches between captions and charts can result in key patterns being overlooked [59]. As with visual annotations, textual design choices may steer attention toward belief-consistent patterns and increase susceptibility to confirmation bias. Together, these findings underscore how the interplay between viewers’ prior beliefs and key visualization design components—such as visual encodings, annotations, and textual elements—can potentially shape data interpretation and contribute to the manifestation of confirmation bias in data visualization.

The goal of my dissertation research is to examine **the role of text, annotations, and visual encodings in shaping confirmation bias during visualization interpretation** for both *static* and *interactive* visualizations by focusing on four research questions:

- **RQ 1: How do people represent internal beliefs using visualization?**
- **RQ 2: How do textual summaries and visual annotations influence confirmation bias in interpreting static visualizations?**
- **RQ 3: How does confirmation bias manifest through user interactions with visualizations?**
- **RQ 4: How can visualization design choices be used to mitigate confirmation bias?**

As depicted in Figure 1.2, we address these four research questions by focusing on three aspects of the process of human reasoning with data visualizations: (1) capturing prior beliefs (RQ1), (2) understanding the effects of visualization design choices (RQ2-3), and (3) mitigating confirmation bias through visual encodings (RQ4). To begin with the first component—human prior beliefs – we examine how individuals naturally represent their prior beliefs internally using visualizations (RQ1), the

elicitation of which is an essential prerequisite for measuring confirmation bias in visual data interpretation. The second component – visualization design choices – encompasses the next two research questions (RQ2, RQ3), which together investigate how different aspects of design impact the manifestation of confirmation bias. In the context of static visualizations, we examine how design choices such as textual summaries and visual annotations influence interpretation, particularly when individuals interpret ambiguous information about polarized topics (RQ2). Building on this, we explore interactive visualizations to understand how interaction – a defining design choice of modern visualization systems – affects reasoning processes and influences the exhibition confirmation bias (RQ3). Specifically, we focus on how users selectively filter or emphasize data points in ways that reinforce patterns consistent with their prior beliefs.

Finally, informed by insights from both static and interactive settings, we propose and evaluate intervention strategies aimed at mitigating confirmation bias (RQ4). These strategies involve manipulations of visual encodings to reduce the salience of belief-congruent patterns when strong prior attitudes are present. Together, these interventions are designed to prompt users to consider disconfirming evidence and, ultimately, foster more objective and balanced decision-making through data visualization.

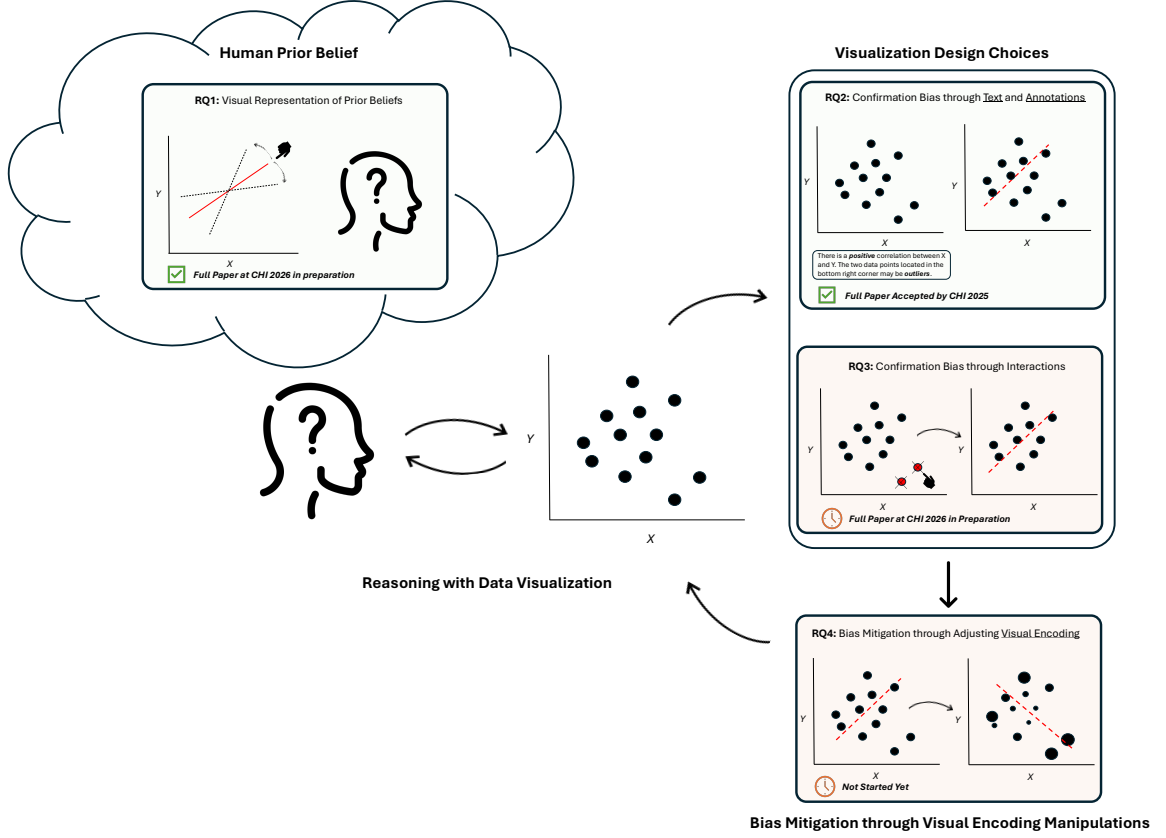


Figure 1.2: This work explores both human prior beliefs and visualization design choices across four research questions. By eliciting prior beliefs more expressively (RQ1), we can more accurately capture individuals' prior beliefs and assess the degree of confirmation bias across visualizations with varying design choices (RQ2, RQ3). In turn, these insights will guide the development and evaluation of interventions aimed at mitigating confirmation bias by strategically adjusting design choices (RQ4), ultimately helping users become aware of their biases and make more informed interpretations of visualizations.

Chapter 2

Related Work

2.1 Cognitive Bias in Visualization

Cognitive bias, systematic but flawed judgments that arise from simple and fast heuristics, can influence how people interpret and make decisions based on data visualizations [136]. Dimara et al. taxonomized 154 cognitive biases into seven categories, detailing how they manifest in different visualization tasks [35]. Furthermore, Wall et al. proposed theoretical computational metrics to characterize five different biases in visual analytic processes [124].

Within these foundational frameworks, several research works have also explored how specific cognitive biases manifest in concrete visualization scenarios. For instance, anchoring bias—the tendency to over-weight initial information when making a decision—has been identified in visual analytic processes. Studies have shown that users examining certain data attributes within the same visual reasoning task are influenced by the initial information presented, such as data descriptions and previously viewed charts [125, 120]. Additionally, Xiong et al. identified the illusion of causality, where participants perceive correlation as causation when interpreting a line chart [137]. Dimara et al. identified the attraction effect in visual reasoning

processes, showing that when participants make decisions between two data points in a scatterplot that are hard to compare, adding a third, clearly worse data point (a decoy) can make them more likely to choose the data point looks better than the decoy [34]. Building on this broader understanding of cognitive biases, this thesis narrows its focus to confirmation bias in visual data interpretation.

2.2 Confirmation Bias

Confirmation bias has been extensively studied in cognitive psychology and is recognized as one of the most influential and persistent forms of bias in human reasoning [88, 40, 132]. It affects how people seek, interpret, and recall information, often leading to skewed decision-making processes that reinforce their existing beliefs. In the following sections, we review foundational theories of confirmation bias and examine its manifestations in visual data interpretation.

2.2.1 Confirmation Bias in Cognitive Science

Confirmation bias is a pervasive cognitive phenomenon in which individuals tend to seek out, interpret, and remember information in ways that reinforce their prior beliefs [88]. It can also be understood as a tendency to preserve, or a reluctance to abandon favored hypotheses [63, 64]. Confirmation bias can influence multiple stages of human hypothesis development [63]. For example, individuals may hold strong prior beliefs that persist even after receiving new evidence, consistent with a Bayesian updating process. They may also revise their confidence in a favored hypothesis insufficiently by downplaying the strength of contradictory evidence, or struggle to generate alternative hypotheses even when they sense limitations in their current one. Confirmation bias can lead to a wide range of potentially adverse consequences. In scientific research, it may cause researchers to favor evidence that supports their

preferred theories, potentially compromising scientific objectivity [103]. In the political domain, confirmation bias can contribute to political attitude polarization, reinforcing societal divisions. In the context of criminal investigations, it can also lead authorities to focus selectively on evidence that confirms a suspect’s guilt or innocence, thereby biasing legal investigations and prosecutions—sometimes resulting in wrongful convictions that permanently alter individuals’ lives [30, 78]. On the other hand, recent work also suggests that confirmation bias may have adaptive functions. For instance, Peters introduced the reality-matching account, proposing that the persistence of confirmation bias is partly due to its social utility. According to this view, confirmation bias enables individuals to affect others’ beliefs and social structures to align with their own, thereby minimizing dissonance with the social world and enabling more effective social navigation [93]. While confirmation bias has been extensively studied in these various domains, it has received relatively little attention in the context of visual data analytics.

2.2.2 Confirmation Bias in Data Visualization

Visualizations are typically designed as an efficient channel for uncovering insights from complex datasets by compactly encoding large quantities of data and making intricate patterns more coherent and accessible [117]. In parallel, the use of visualization can also make it easier for readers to identify visual patterns that align with their theories and, thereby reinforce confirmation bias by interpreting supportive visual patterns as the primary takeaway. For example, prior research has shown that individuals’ prior beliefs influence how they perceive the strength and direction of correlations in scatterplots [140, 141]. To address this bias, researchers have proposed mitigation strategies such as visually annotating contradictory data patterns to promote more balanced reasoning [25, 31]. In addition to visual content, textual elements in visualization designs – such as data summaries, captions, and title, also

play a significant role in shaping the interpretation of the data the same as these textual contents and can also act as strong triggers of confirmation bias beyond visual contents in visualization designs [65, 59, 6]. Motivated by the influence of both visual and textual content on visualization interpretation, this thesis investigates the extent to which textual content, annotations, and visual encoding choices influence the degree of confirmation bias in visual data reasoning. These findings motivate us to move beyond simply detecting confirmation bias in visual reasoning, towards quantifying its strength and examining how design choices in visualization could influence the extent to which this bias is exacerbated or mitigated.

2.3 The Impact of Encodings on Visual Data Interpretation

The choice of visual encodings plays a critical role in facilitating readers’ ability to accurately interpret data values from visualizations [100, 26]. For example, when making proportional comparisons, readers can make more accurate estimations using pie charts compared to bar charts [106], and are better at interpreting linear correlations in scatterplots than in parallel coordinate plots [74]. One of the most frequently cited studies on the effectiveness of visual encodings is by Cleveland and McGill [26], who asked participants to compare values and estimate percentage differences using various encodings—such as positions along an axis, the length of line segments, the area of circles, the curvature of lines, and angular measures. Based on participants’ estimation accuracy, they established a ranking of encoding types in terms of perceptual effectiveness. Their findings emphasized that visualizations should prioritize encodings that are easier for viewers to interpret—such as position and length—over those that are harder to judge, like area or color.

As visualizations incorporate more dimensions of data into a two-dimensional

space, additional visual encodings are often used to represent supplementary variables in interactive visualization systems [86, 87, 108]. For example, in scatterplots, attributes such as size, opacity are commonly employed to encode extra information [84]. However, these additional encodings—particularly size and opacity—may interfere with the viewer’s ability to accurately perform one of the scatterplot’s core tasks: interpreting the linear relationship between the x and y variables. In two recent studies, Strain et al. [113, 112] demonstrated that systematically increasing the size and opacity of data points with smaller residuals can bias users’ perception, leading them to overestimate the strength of the correlation compared to scatterplots without such manipulations. These findings inspire us to consider such visual encoding manipulations as a double-edged sword: while they can be used to mislead or reinforce biased interpretations, they may also be repurposed – when users’ prior beliefs are known through elicitation – to downplay belief-consistent correlations in scatterplots and encourage a more balanced reasoning process.

2.4 The Impact of Text and Annotations on Visual Data Interpretation

A growing body of research has shown that both textual and visual annotations embedded in data visualizations significantly shape how viewers interpret, recall, and reason about information. Regarding textual content, readers often prefer its integration with visualizations to enhance comprehension [110], but this combination can also introduce interpretive biases. For example, prior work has demonstrated that biased titles can disproportionately influence what viewers remember—sometimes more so than the graphic content itself [66, 65]. Similarly, captions that align with salient visual features tend to guide viewer takeaway, while mismatches between captions and visual content can result in overlooked information [59].

These effects are amplified when textual elements interact with viewers’ prior beliefs. Kong et al.[66] found that titles inconsistent with an individual’s beliefs can reduce the perceived credibility of a visualization—even if the visual data itself aligns with those beliefs. Likewise, Stokes et al.[111] showed that the presence of textual annotations can increase perceived author bias, especially when those annotations contradict the viewer’s existing attitudes.

Beyond text, visual annotations also play a critical role in shaping interpretation. While early perspectives emphasized reducing chartjunk to maximize clarity and objectivity [117, 116], more recent research has challenged this view by showing that annotations can improve comprehension and recall [5, 55]. Crucially, annotations do more than guide attention—they can actively shape interpretation. Recent work has demonstrated that annotated visuals lead viewers to draw conclusions aligned with the annotation, even when the same visual evidence is available without annotation [6, 111].

Together, these findings highlight how textual descriptions and visual annotations act as interpretive lenses through which data is filtered. When aligned with a viewer’s prior beliefs, they can reinforce belief-consistent narratives and increase susceptibility to confirmation bias. As such, visualization designers must be cautious about how text and visual embellishments are combined – an issue we further investigate in Research Question 2 of this thesis.

2.5 Belief Elicitation in Visualization

Confirmation bias reflects nuanced interactions between individuals’ prior beliefs and evidence presented to them. Thus, accurately eliciting prior beliefs is essential for measuring confirmation bias, as it enables a more precise comparison between how evidence is interpreted and the beliefs that shape that interpretation. Additionally,

the study designs in this dissertation primarily center on people’s prior beliefs about a dataset topic and assess how they interpret evidence presented through visualizations of that dataset. Accordingly, we review prior work on visual belief elicitation techniques that map people’s prior beliefs to the visual data they are shown.

Researchers in the visualization community are increasingly focused on designing more effective elicitation techniques by incorporating visual representations to enhance the expressiveness of the elicitation process. The VIBE framework [82] envisioned a design space for visual belief elicitation in the context of data journalism, offering high-level considerations for designing elicitation strategies based on user characteristics such as graphical literacy, statistical literacy, and domain expertise. VIBE aimed to provide tailored elicitation designs that cater to the diverse needs of users, ensuring engagement and comprehension.

Building on this framework, Koonchanok et al. demonstrated that participants who used a graphical slider to express their prior beliefs about the strength of relationships between variables were more successful in making accurate inferences about the relationships presented in subsequent data visualizations [68]. Similarly, Karduni et al. assessed a visual elicitation technique “line + cone” designed to capture beliefs about linear correlations and their uncertainty through a set of slopes [58]. In another recent paper, Koonchanok et al. introduced an innovative exploratory visual analytic system, which encouraged users to articulate their beliefs by sketching expected trend lines prior to encountering new data, thereby prompting users to scrutinize disparities between the incoming data and their preexisting beliefs [67].

Collectively, these efforts provide specific elicitation techniques and also inform the design of techniques that are responsive to the form in which evidence is presented – particularly through visualization.

Chapter 3

Characterizing Internal Representations of Attitudes and Beliefs

Understanding and measuring confirmation bias begins with accurately eliciting human prior beliefs or attitudes. However, to do so effectively, it is crucial to first understand how individuals internally represent beliefs and attitudes. This foundational insight could inform the design of elicitation techniques that better align with people’s internal representations of these abstract concepts, ultimately enabling more accurate and meaningful assessments of confirmation bias. This chapter presents the work conducted in response to **Research Question 1**, which is currently being prepared for submission to ACM CHI 2026.

RQ 1 *How do people represent internal beliefs using visualization?*

Belief elicitation techniques span a spectrum from expressive to structured methods. *Expressive* elicitation techniques, such as open-ended text input from think-aloud protocols and interviews [89, 46], rely on verbal responses to provide rich insights into

user attitudes and beliefs. However, while expressive methods capture detailed perspectives, they can be difficult to analyze systematically. To improve analyzability, *structured* elicitation methods, such as Likert scales and sliders streamline analysis for researchers while offering users a more efficient and intuitive response format [14]. Yet, this efficiency also comes at a cost—pre-coded response options may constrain expressiveness, making it difficult to capture nuanced attitudes and beliefs. Thus, the choice of elicitation method involves a trade-off between analyzability and expressiveness.

To address this trade-off, researchers have increasingly explored **visual elicitation techniques**—methods that engage users in expressing beliefs through interaction with visualizations. For instance, to support better reasoning with proportional data, Kim et al. elicited participants’ perceived probability distributions by having them distribute a set number of balls into bins [61]. Similarly, Heyer et al. used an interactive bar chart with adjustable heights to elicit users’ prior beliefs as percentages. Their findings suggest that externalizing prior beliefs in this way enhances participants’ ability to acquire new information from data-driven storytelling by highlighting the deviation between their prior beliefs and the actual data [52]. More recently, Karduni et al., utilized lines with slopes that could be adjusted by dragging to elicit prior beliefs as the relationship between two variables [58]. Despite these advancements, many of these techniques have not considered how people mentally represent their attitudes and beliefs, thus begging the question: how well can these visual elicitation techniques capture an individual’s prior attitudes and beliefs?

To bridge this understanding and inform the design of future visual elicitation techniques that better align with how individuals internally represent their attitudes and beliefs, we turn to the most expressive form of visual elicitation: free-form drawing, a visual analogue to open-ended text responses. We provided 41 participants with a canvas-based interface to externalize their attitudes and beliefs through drawing, enabling us to gain insight into the strategies they used to visually express these

internal representations. To complement this, we also incorporate the textual channel as natural language to allow us to interpret their drawings, and for participants to clarify and supplement details that may be otherwise constrained by their drawing abilities.

Using thematic analysis of the resulting drawings and text responses, we identified several strategies that users employ to visualize attitudes and beliefs from basic elements such as faces and symbols to more complex forms such as stories, diagrams, or equations. Building on these naturally occurring visual strategies, we propose alternative visual elicitation techniques that expand expressive possibilities and emphasize the need for future research to validate and refine these methods. Beyond elicitation methodologies, our findings provide valuable insights for researchers and professionals developing survey instruments to better understand diverse populations across a variety of topics.

3.1 Qualitative Study Methodology

Study Motivation. Cognitive psychologists have long been debating how people represent information internally, led by two prominent figures in this area: Stephen Kosslyn and Zenon Pylyshyn. Kosslyn and colleagues believe that information is spatial and depictive, therefore pictorial in nature [69], while Pylyshyn and colleagues argue that information is propositional, language-like, symbolic internal representations [92, 94]. While this debate is far from being conclusive, more recent visualization research has emphasized the equal importance of text and visualization in information communication[50]. This research further highlights the preferences of a minority who favor text-only information, underscoring the importance of acknowledging text as a standalone factor in the assessment of data visualization designs.

In light of this discussion and recognizing the importance of both visual and verbal

Table 3.1: The **Domains** and **Prompts** assigned to participants across Round 1 and Round 2.

Round	Domain	Prompt
1	Vaccination	“What are your attitudes towards vaccination?”
	Education	“What are your views on the relationship between education spending and academic achievements?”
2	Generative AI	“What are your attitudes towards generative AI?”
	Transportation	“What are your views on the relationship between the use of bicycle and the occurrence of traffic accidents?”

mediums for communication, we conducted a two-round qualitative study to understand how people mentally represent attitudes and beliefs. We provided a free-form digital drawing canvas on a tablet device to enable participants’ visual expressions, accompanied by open-ended verbal responses in natural language to enrich and contextualize these visual expressions. The first round focused on collecting a diverse range of visual and verbal representations of attitudes and beliefs to develop a comprehensive codebook. The second round, following the same structure as the first but using different topics to elicit responses. It was conducted to validate the robustness of the existing codebook and to identify any emerging patterns in visual and verbal responses that could inform the addition of new codes. Our goal was to understand the characteristics and preferences among visual and verbal representations of attitudes and beliefs.

Participants. We recruited $N_1 = 20$ participants in the first round (10 women and 10 men) and $N_2 = 21$ participants in the second round (11 women, 9 men and 1 non-binary individual) from the crowdsourcing platform Prolific [90]. Our eligibility criteria included participants who were at least 18 years old, located in the United States, fluent in English, and able to complete the study using a tablet device, as required by the study protocol. Participants who completed the first round were ineligible to complete the second round. Studies were conducted via Qualtrics [105]. Across the two rounds, we paid a flat rate of \$3.35 to complete the study, which took median times of 17.62 and 22.02 minutes respectively.

Study Design. Both rounds of our study adopted a within-subject design where

each participant was asked to express their attitudes or beliefs about two different topics. In the first round, participants were asked to express their attitudes towards “Vaccination” and beliefs on “Education.” Based on the results of the first round, we derived a code book that captured the key themes in the responses. The second round followed the same procedure as the first round to examine its robustness and identify emerging patterns across two new topics: attitudes towards “Generative AI” (Artificial Intelligence) and beliefs on “Transportation.” The phrasing used to prompt user attitudes and beliefs in the two rounds is summarized in Table 3.1.

We operationalize attitude as the perceived likelihood that something is viewed as positive or negative, and belief as the perceived likelihood that certain attributes are associated with the topic or object of discussion [143]. We chose four topics where participants are likely to hold strong opinions, aiming to elicit more pronounced attitudes and beliefs. Rather than analyzing how attitudes and beliefs are distributed across each topic, our focus is exploring the visual representations participants use and how these visuals convey their perspectives. To achieve this, we intentionally framed our instructions to be vague to avoid biasing participants, allowing for a wider range of user interpretations to obtain a broader sample of responses.

Using drawing to gather perceptions on attitudes and beliefs. To inform the design of future visual elicitation techniques, we employ digital drawings with a canvas interface on a tablet as an open-ended technique that enables participants to visually express their attitudes and beliefs in a free-form manner. This approach allows us to access these nuanced visual representations and synthesize common patterns across individuals, providing valuable inspiration for the elements and structures of future visual elicitation techniques.

Drawing is more often used in research studies involving children, as researchers have been concerned that cultural standing and willingness to engage may significantly affect adult drawings[81]. However, drawing remains an effective channel for

articulating perceptions and can be greeted with pleasure among adults as well [12]. Research studies have used drawing to investigate differing opinions on abstract concepts (e.g., what is energy, what is information)[48, 49, 12]. These studies found that drawing returns participants to their youth, inspiring beautiful and insightful drawings for compositional interpretation and thematic analysis. This approach explores how concepts are culturally constructed, such as when participants drew light bulbs, sockets, and batteries to represent "energy," connecting the concept to more than just heat or electricity. Therefore, to thoroughly explore the mental constructs of attitudes and beliefs, we use drawing to connect participants to their social and cultural worlds, accessing their tacit knowledge and unconscious perceptions [102, 81].

Table 3.2: The instructions participants were given for visual and verbal conditions in both studies.

Condition	Prompt
Visual	"Please draw <i>a picture</i> to represent your views either using the canvas below, OR drawing your picture on a piece of paper."
Verbal	"Please type <i>a description</i> to represent your views in the text entry box below."

Conditions: Comparing Visual and Verbal Attitudes/Beliefs. To compare visual and verbal responses, participants expressed their attitudes or beliefs about each topic by drawing (visual) and typing (verbal). For the verbal response, participants were provided an essay text box in Qualtrics, with a character limit set at 20,000 (participants provided responses ranging from $min = 5, max = 2146, \mu = 226.62$ characters). For the visual response, participants were provided a 1000×600 pixels canvas with basic drawing tools to adjust pen size (from 1 to 72 pixels), a color picker, and an eraser tool. Participants could use this canvas to draw a response or draw elsewhere and upload an image file. The language used in the visual and verbal conditions is summarized in Table 3.2.

Procedure. After obtaining informed consent, participants engaged in a warm-up activity where they were asked to (i) write about and (ii) sketch their favorite fruit. Participants then completed visual and verbal tasks for two topics in each

round, expressing their attitudes or beliefs about each topic both visually and verbally, with the order of modalities and topics counterbalanced. For both response types, participants were also asked to provide comments on their answers to help the authors better understand their verbal and visual responses. After completing the two tasks for each topic, participants reported their preference for expressing their attitudes and beliefs via the visual versus verbal response, and selected the modality they felt more accurately capture their attitudes and beliefs.

Analysis Methodology. After the first round of the study, two authors independently conducted a thematic analysis [13] to derive a preliminary set of codes each for the verbal and visual responses. The resulting codes had significant overlap and were ultimately combined together to produce a codebook after discussion with all authors. Two different authors then independently coded the results from the first round, achieving an inter-rater reliability using Cohen’s kappa [27] of 0.79, indicating moderate agreement. Subsequently, all disagreements were discussed and resolved to achieve consensus labels.

After the second round, the authors discussed the disagreements and introduced new codes to better capture nuances in the data. These included verbal codes for personal experiences (using personal experiences to rationalize a belief) and conditional beliefs (depict a belief under specific condition). The data from both rounds were re-coded using the updated codebook. For the second round, The two authors achieved a final inter-rater reliability score of 0.44, indicating weak agreement. The lower inter-rater reliability resulted from two factors: (1) different interpretations of the code belief in verbal responses, and (2) the introduction of new data and emerging verbal codes in round 2, which required further adjustments to the codebook. Specifically, the belief code was intended to capture expressions of belief only for questions explicitly asking about beliefs (Topic “Vaccination”, “Generative AI”). However, one author also applied this code to rationales for attitudes in attitude-related questions (Topic

“Education”, “Transportation”). All disagreements were subsequently discussed and resolved. The final codebook and coded data are included in the supplemental materials.

3.2 Findings

3.2.1 Summarizing the Code Book

We first summarize the final code book used in the two rounds for visual (Table 3.3) and the complementary verbal responses (Table 3.4). In the forthcoming sections, we will discuss data across both rounds together. Both code books include a description of each code and its prevalence, as well as an example. The codes are not mutually exclusive, i.e., a response can fit multiple codes. When responses (text or drawings) were ambiguous, we utilized the optional explanatory comments provided by participants at the end of the task to disambiguate our interpretation.

3.2.2 Visually: *What* attitudes/beliefs do people convey?

We present an analysis of visual attitudes/beliefs types observed from participants’ sketches. These include directional attitudes, attitudes through expressions, beliefs through structural objects, uncertainty, and topics, described below with examples.

 **Directional Attitudes (15).** Fifteen sketches were found with attitudes conveyed with a directional stance, which can be positive, negative, or neutral. Participants frequently incorporated symbols or verbal elements in their sketches, such as using ✓ marks or ✗ marks to indicate positive or negative attitudes. One participant used the hand-sketched word “yes!” to signify a positive attitude.

 **Attitudes through Expressions (27).** We found 27 sketches where participants drew facial expressions to capture attitudes. As one of the most powerful non-verbal communication channels, facial expressions can convey a

wide array of emotions and intentions [37]. For example smiling faces can indicate support for a topic and fearful faces can convey concerns. One notable case (P27) involved the use of a smiley face with a frown to communicate mixed feelings about vaccination, acknowledging its efficacy while also hinting at the potential discomfort involved.



Beliefs through Structural Objects (37). We observed participants

utilizing structured objects to visualize their attitudes and beliefs. For instance, when asked about the usage of bicycles and traffic accidents, P41 illustrated their belief towards an eventual accident-free coexistence by drawing a bicycle and a car each in its lane. Another participant, P36, highlighted their belief in the current limitations of AI applications by displaying a sign that read “limited AI” beneath an exclamation mark. Additionally, we noted that depictions of beliefs often fell into either structural visual representations such as schematic diagrams or narrative scenarios, akin to stories featuring human interactions, and occasionally, hand-drawn dialogues, which we elaborate on in subsection 3.2.3.



Uncertainty (3). In practice, beliefs are seldom certain [24, 114]. Com-

peting results from scientific findings, unfamiliarity, forgetfulness, and changing levels of interest in the topic can make our beliefs noisy. For example, someone without much prior exposure to a topic might not have a strong belief in its trends and can be more easily swayed. Some participants displayed a lack of familiarity with the question topic (e.g., “Generative AI”) and depicted attitude uncertainty in their sketches, either through symbols or facial expressions. For example, P21 used a question mark to signify their lack of experience with Generative AI, while P16 and P22 used facial expressions featuring a subtle downward curve of the mouth to indicate their uncertainty towards “Vaccination” and “Generative AI” respectively.



Topics only (11). Eleven sketches depicted the topic without providing

any insight into the user’s attitudes or beliefs. We labeled them as topics only.

One example from P19 solely includes the hand-sketched word “education” to depict the subject. We also observed a drawing of a syringe to symbolize “Vaccination” (P10) and a bicycle for “Transportation” (P34). However, upon examining the associated verbal responses, it was evident that participants held personal beliefs not captured by their drawings. The participant who sketched the word “Education” (P19) elaborated in their verbal response, stating *“I firmly believe that allocating sufficient resources and funding to education can greatly impact the quality of learning and overall academic performance.”* This revealed their beliefs as a strong positive relationship between education spending and academic achievements.

3.2.3 Visually: *How* do people convey attitudes/beliefs?

Next, we share how participants combined the aforementioned visual elements to articulate varying attitudes and beliefs.

3.2.3.1 Simple Compositions (36)

Some participants communicated their visual attitudes or beliefs using a combination of text, symbols, and pictorial representations, which we refer to as ‘simple compositions’. These simple compositions may represent a single subject (22/36) or multiple subjects (14/36).

 **Single Object-Attitude induced drawings (11/22).** The most direct method to convey attitudes is through depicting a face (8/11), symbol (2/11), or sketched text (1/11), which directly conveys an attitude towards the topic in question. For example, a smiley or angry face, an ✕ symbol or tick mark, or a thumbs up symbol.

 **Single Object-Ambiguous pictorials (11/22).** Another form of single-subject sketch uses a pictorial symbol that conveys meaning through the resemblance to a physical object to represent the topic without any attitude or

Table 3.3: The Final Codebook: Visual Codes.

Category	Code	Description	Example	Prevalence (#/82)
Object	Pictorals	Depicts items that are not people or faces		60
	Face	Depicts a face, likely showing emotion		31
	Posture	Depicts the posture of a human body		26
	Text	Depicts words written or typed out		23
	Symbols	Depicts symbols such as hearts, stars, or X's		14
	Visualization	Depicts a chart or other form of abstract data visualization		4
Composition	Story	Depicts a scene that indicates relationships between objects		25
	Diagram	Depicts a schematic diagram, possibly using arrows to indicate relationships; may suggest a causal relationship		8
	Superimposition	Depicts containment or modification of an object to change its meaning (e.g. an object enclosed in a crossed out circle)		6
	Equation	Depicts a mathematical relationship between entities		6
Miscellaneous	Conveys Attitude	Conveys attitude or opinion towards a topic		46
	Topics Only	May describe the topic, but does not indicate the user's attitude/beliefs towards it		11
	Unintelligible	The drawing cannot be deciphered on its own due to complexity, penmanship, etc.		4
	Non-traditional	Indicates non-traditional responses such as searching the web for images rather than drawing them		1

belief. For instance, some participants only drew a syringe (P10) or a robot (P34, P40) when asked about their attitudes towards “Vaccination” or “Generative AI”.

This tendency could be attributed to the participants’ neutral viewpoints, misunderstanding that the drawings were meant to depict their attitudes or beliefs, or even their aversion to drawing (8).

 **Multiple Objects- Juxtaposition with symbols (8/14).** The first multi-subject strategy we noted involves both representations of topics and attitude as a juxtaposition: topical pictorials and symbols depicting attitudes or beliefs. Participants using this method typically choose one item to symbolize the main topic and then employ either a facial expression (6/8) or a symbol (2/8) to communicate their stance. For example, P3 drew a smiley face next to a syringe to indicate a positive attitude towards “Vaccination”, while P38 used a thumbs-up symbol alongside a laptop to signify a positive attitude towards “Generative AI”.

 **Multi Objects- Superimposition of pictorials (6/14).** The other multi-subject strategy we observed involved the utilization of multiple graphical objects superimposed on each other (6), which might be influenced by a metaphor of street signage where participants compose symbol and attitude together to express their stance on the topics. Most of the superimposition responses appeared in visual attitudes (5). For instance, we observed drawings of a syringe enclosed within a heart shape (P9) or a crossed-out circle symbol (P6) to signify positive or negative attitudes towards “Vaccination”. The remaining one (P29), interestingly, represented containment with “30mph” sketched inside, depicting the user’s conditional belief that only higher overall traffic speeds could lead to an increase in accidents if more bicycles are present.

3.2.3.2 Complex Compositions (46):

Through tangible stories, diagrams, equations, and data visualization, participants created structured compositions that allowed them to communicate more complex attitudes and beliefs.

Stories (28/46). *Story* is the most commonly employed composition form of sketch to visualize participants’ attitudes or beliefs about a prompt in a way that’s tangible and vivid to them. Different from multi-subject strategies, a story typically depicts a scene to represent the interactions between objects related to the topic.



For instance, P20 illustrated a relieved patient finally receiving protection from COVID-19 by getting vaccinated, thereby expressing a positive attitude toward vaccination. P41 depicted a car and a bicycle each in

their respective lanes, symbolizing the belief that adherence to lane discipline can prevent traffic accidents. The story form of composition is mostly employed (14/28) to depict beliefs as relationships between the use of bicycles and the occurrence of traffic accidents. This may be because transportation plays a role in people’s everyday lives and is therefore natural to mentally represent as a story. Participants used a combination of facial expressions (8/25), symbols (3/25, e.g., a ✓ symbol), and sketched text (5/25, e.g., a hand-sketched word “yes”) in their stories to convey attitudes towards the topics. This is different from only using pictorial representations of the topic without incorporating attitudes to formulate a story.

Abstract compositions (18/46). Participants created abstract-format sketches such as diagrams, equations, or data visualizations to represent the relationships between various variables.



Diagrams (8/18). Unlike stories, diagrams depict schematics and are employed, often with arrows, to illustrate causal relationships or causes for attitudes or beliefs. For instance, P20 sketched a green dollar sign connected by an arrow to a smiling student wearing a graduation cap. This

illustration suggests their belief of a positive correlation between education spending and academic achievements. P26 drew a smiley face linked by an arrow to a surprised face, signaling their current satisfaction with AI while underscoring future uncertainty. Overall, Diagrams are primarily used in response to the questions asking

for beliefs as relationships (5/8). The majority (5/8) of those who used diagrams did not consider drawing to play a role in their lives. These findings could imply that diagrams may serve as an intuitive and efficient way to represent beliefs about causal relationships. Other diagram authors conveyed attitudes by also using faces (3/8) or sketched text (1/8). Several participants used topic-related objects within their diagrams to illustrate specific beliefs and explained their attitudes/beliefs in verbal responses. This observation suggests a natural inclination to express attitudes even when not explicitly asked to do so.

Equations (6/18). We categorized equations, depicted in six sketches, as an abstract composition strategy to present attitudes and beliefs using mathematical relationships. The majority of the equations (4/6) communicated beliefs; participants tended to construct complex equations that incorporated multiple factors related to their beliefs. For instance, P15 illustrated "woman in dress + cash + computers," linked by an = sign and finally pointing to a doctoral hat, to signify that education spending can contribute to academic achievement but also requires good teachers and educational facilities. The remaining two sketches (P7, P19), expressing attitudes toward vaccination, were more concise and direct. Both sketches used symbols like a syringe and a ✓, connected by an = sign, to convey a favorable stance on vaccination.

Data visualization (4/18). Participants also used abstract data visualization techniques to represent complex beliefs. One participant (P13) used a bar chart, while the other three (P9, P11, P39) used line charts. Three out of the four participants stated that drawing plays no role in their lives. Although three participants still favored verbal responses, believing them to better align with their mental models of beliefs, data visualization served as an effective way to illustrate beliefs on complex causal relationships.

Table 3.4: The Final Code Book: Verbal Codes.

Category	Code	Description	Example	Prevalence (#/82)
Expressions of views	Attitude	Conveys attitude or opinion towards a topic	<i>"I'm against any type of vaccination"</i>	41
	Belief	Conveys a belief in a trend or relationship	<i>"I believe money does help you be better educated."</i>	37
	Hedging	Describes caveats to the belief, often includes the use of the word "but" to affirm exceptions	<i>"I think vaccination is necessary for some illnesses. But I do not think babies should be over-vaccinated."</i>	37
	Math	Uses language around mathematical relationships such as plus, equals, linear, etc.	<i>"More dollars = more achievement."</i>	16
	Relationship	Describes a relationship between two concepts or entities, often causal in nature	<i>"I feel the more they invested in education spending, the more achievements will be achieved"</i>	11
	Uncertainty	Indicates lack of knowledge or sense of unpredictability	<i>"I have never used generative AI"</i>	10
	Conditional belief	Describes a relationship between two concepts or entities, often causal in nature under specific conditions	<i>"When rules are not followed by automobiles on the road and bicycles accidents happen."</i>	9
Reasons for views	Figures of Speech	Uses metaphors, analogies, or colloquial expressions	<i>"I think it (Generative AI) can be a big timesaver and make jobs more automated."</i>	8
	Rationale	Provide details or rationale, often includes the use of word "because" to provide explications	<i>"The use of bicycle reduce traffic accidents because less people are on the road."</i>	32
	Speculation	Speculates reasons for a relationship	<i>"Education spending is likely to provide the latest technological advances for students"</i>	18
	Morality	Invokes morality; may use phrases such as "should" or "should not", "right" or "wrong", etc.	<i>"I think it (Generative AI) is terrible for humanity in general. It has no real benefits and should be banned."</i>	8
	Emotion	Indicates an emotional response to the topic such as happiness, fear, or disgust	<i>"I think Generative AI is strange and unknown to me so i'm a little bit afraid of it."</i>	6
Evidence / Logic	Personal experience	May use personal experiences to make a point	<i>"In my city, I do not think bikes have an effect on traffic accidents."</i>	4
	Data	Mentions references to historical data or past studies	<i>"They have a proven track record of preventing the spread of infectious diseases"</i>	1
Outcomes	Wishes	Describes a future preference	<i>"I think people should be more careful with bicyclists because they are normally hard to see."</i>	8

3.2.4 Verbally: *What* attitudes/beliefs do people convey?

In order to enrich our interpretation of the visual data and uncover attitudes and beliefs that may not have been fully captured due to limitations in participants' drawing abilities, we applied the same thematic analysis to their verbal responses. Similar mental constructs of attitudes and beliefs emerged from the verbal responses as from the visual responses. Most attitudes (41/41) and beliefs (27/41) were clearly articulated through verbal responses. An illustrative example comes from a participant who wrote, *"Vaccination is crucial for controlling and/or eliminating severe diseases."*

Verbal responses allowed participants to express *complex* attitudes and beliefs. For instance, hedging was a recurrent feature observed in participants' verbal responses (17/82). Hedging characterizes responses that introduced uncertainty, caveats, or exceptions to attitudes and beliefs. P4 provided a telling example: *"I suppose the more spent on the infrastructure, the more opportunity exists for students to achieve academically. But correlation doesn't imply causation."* This nuanced stance clarified that the participant did not intend their beliefs to be construed as causation.

Interestingly, the study identified various visual forms of attitudes and beliefs as indicators of hedging. Symbols like $\approx$ signaled a distinction between correlation and causation in the Education topic. Faces combining smiles and frowns depicted mixed feelings, such as cautious optimism about Generative AI. Furthermore, the analysis of verbal responses revealed that the content of participants’ sketches varied based on the importance of their hedged concerns. For significant issues depicted in the verbal response such as P1’s statement: *“I think vaccination is necessary for some illnesses. But I do not think babies should be over-vaccinated.”*, the corresponding visual response depicted detailed imagery including a crying baby alongside a syringe—to only convey concern about over-vaccinating infants, even though the verbal response reflected partially positive attitude towards vaccination. Conversely, for minor concerns within the verbal responses (P35: *“I think it’s interesting and useful, but for now isn’t good enough to have lots of uses.”*), participants portrayed their overall stance, like using a thumbs-up to signify general positivity towards Generative AI despite recognizing its limitations.

3.2.5 Verbally: *How* do people convey attitudes/beliefs?

The most common method of expressing attitudes and beliefs observed in the verbal responses was rationale (logical basis or justification) for attitudes (18/82) and beliefs (13/82), largely through speculations (14/82, forming a conjecture without firm evidence). For instance, P30 provided a rationale to their belief about the relationship between bicycle use and traffic incidents, stating, *“I think bike riders do not pay enough attention and cause accidents by thinking they rule the road.”* We also found that participants often preferred expressing more complex rationale through text rather than visuals. For example, P33 described, *“Generative AI is mostly a bad thing since it uses public data without getting the consent of the person that put the data online. It also could end up giving bad advice or information that could have*

disastrous effects on the technology that relies on them. They are also still in their infancy stages and are undergoing updates and could cause more harm than good. The fact more people are becoming reliant on technology like chat bots is also not a good thing since it means that less people will be knowledgeable and will rely too heavily on chatbots which are not foolproof nor 100% reliable.” This individual began by describing data privacy concerns using generative AI and then transitioned to speculations about the broader risks of over-relying on still-evolving technology. Such in-depth perspectives can be challenging to depict solely via sketches. This case illustrates the limitations of sketches in fully reflecting complex thought processes, and how verbal responses can bridge this gap by providing a richer context. Therefore, we consider text alongside sketches for deeper insights. In this example, we can infer the participant’s primary concern: the outweighing of risks over benefits with the advent of Generative AI. This stance could theoretically be symbolized by a simple Equation, “ $Risks > Benefits$,” offering a glimpse into the participant’s visual beliefs.

We also observed the use of mathematical language in verbal responses. For example, some participants mentioned likelihoods (e.g., P15: “*The higher the number of people vaccinated, the **more likely** that a given population will develop herd immunity to serious diseases.*”), and some described correlation (e.g., P32: “*I believe that there is a **positively correlating relationship** between use of bicycles and occurrence of traffic accidents.*”) in their response.

Furthermore, we noticed that participants used figures of speech (16/82), including metaphors (e.g., P3: “*Schools **shape and mold** the future generations.*”; P30: “*I think bike riders do not pay enough attention and cause accidents by thinking they **rule the road.***”) and analogies (P41: “*Generative AI is a **new era.***”)

Additionally, a pattern emerged wherein participants often concluded their verbal responses by expressing their wishes for the future (8/82). For instance, P28 wrote, “*We as a society need to find a better way to incorporate people who ride bicycles*

because as it stands now they pretty much get in the way.” This observation suggests a valuable additional mental construct of beliefs that visual elicitation techniques can explore: allowing participants to separately articulate their *future expectations* as part of their feedback, distinct from the elicited beliefs. For instance, in discussing bicycle use and its impact on traffic, participants could visually depict their concern by placing a bicycle close to a car to symbolize a concerning distance as their *current belief* and then place another bicycle at a safer distance to represent their *future expectation*.

3.3 Discussion

3.3.1 Visual Representations of Attitudes and Beliefs

Our exploration revealed strategies that individuals naturally use to express their attitudes and beliefs visually. Specifically, *attitudes* were often complex and most frequently conveyed through **facial expressions**, occasionally accompanied by symbols or brief text, to reflect support, concern, or ambivalence. In contrast, *beliefs* were structural and most frequently expressed through vivid visual stories. These story-based sketches depicted scenes, interactions, or scenarios that helped participants communicate complex relationships. Some participants also conveyed more abstract beliefs, such as diagrams or equations, to highlight specific **causal relationships**. Within these abstract visuals, participants commonly used pictorial symbols to represent the topic objects, arrows, or = signs to illustrate the directions of relationships, and occasionally employed symbols like $\neq$ sign to explicitly indicate the absence of a relationship.

3.3.2 Going Beyond Verbal Preferences

Participants consistently turned to the verbal channel to articulate complex attitudes and beliefs with precision and ease. From our two-round qualitative study, we found that over 80% of the participants favored verbal responses, believing that this channel most accurately reflects their mental constructs of attitudes and beliefs. Motivated by this strong preference, we revisited the verbal responses to identify aspects of attitudes and beliefs that were more readily expressed through language but omitted or underrepresented in the visual responses. Our goal was to uncover the strengths of verbal articulation and explore how these insights could inform the design of more effective visual elicitation methods.

Internal representations of attitudes and beliefs are often complex and influenced by multiple factors [80, 99, 43]. Through verbal responses, we identified two common **layered structures** of attitudes and beliefs that were often missing from participants' visual representations: attitudes articulated through hedging and beliefs conveyed through conditional reasoning. For attitudes, as discussed in section 3.2.4, hedging was a frequent strategy used to convey nuance, appearing in 17 out of 84 verbal responses. However, 7 of the corresponding visual responses only partially represented the hedged attitude. For instance, participant P28 stated, *“AI is simply a new tool that we created in order to benefit us; it can have both negative and positive effects on society,”* but drew only a smiling robot face—highlighting the positive attitude while omitting the expressed ambivalence. For beliefs, layered structures often took the form of conditional statements that describe relationships between concepts under specific circumstances. Among 9 verbal responses conveying conditional beliefs, 7 expressed the condition only verbally, not visually. For example, participant P39 wrote, *“Traffic accidents will first increase along with bicycle numbers due to more chaotic. But then it will decrease when the average speed went down.”* Yet the accompanying visual response depicted only the trend of accidents rising and falling, omitting the

speed-related condition that contextualized the belief. These observations motivate us to consider the development of visual elicitation techniques that better support the expression of layered reasoning commonly present in verbal articulations.

3.3.3 How Might the Future Visual Elicitation Techniques Look?

Building on the expressive patterns and gaps identified in participants’ visual and verbal responses, we now consider how these insights can inform the design of future visual elicitation techniques. The findings from our qualitative study suggest that while participants readily used visual forms like facial expressions, symbols, and causal diagrams, these often lacked the layered complexity present in their verbal articulations.

We approach these observations not as fixed templates but as a way to imagine what alternative visual elicitation techniques could look like. Our goal is to translate participants’ intuitive visual strategies into structured techniques that are both expressive and analyzable. In doing so, we explore three directions that emerged as both prevalent in participant responses and promising for design:

- First, given the widespread use of facial expressions to convey affective attitudes, we propose using face-based visual elements to elicit **emotional attitudes**.
- Second, we view the absence of hedging and layered nuance in many visual responses not as a limitation, but as an opportunity to expand the expressive potential of visual elicitation. To address this gap, we propose multi-slider interfaces that draw on the strengths of verbal articulation, enabling participants to more precisely express **hedged attitudes** in a structured and analyzable form.
- Third, building on participants’ use of arrows, symbols, and occasional equations to represent cause-effect relationships, we explore using diagrammatic structures to capture **causal beliefs**.

While the techniques we propose are grounded in naturally occurring participant responses, they should be further evaluated for expressiveness and accuracy.

3.3.3.1 Using faces to elicit emotional attitudes.

As one of the most powerful channels of non-verbal communication, facial expressions are central to conveying a wide array of emotions and intentions [37]. The human brain has even developed specialized regions solely dedicated to processing facial expressions [107]. Studies show that faces not only communicate basic emotions but also convey complex social and moral information [115]. Similarly, facial expressions represented as emojis in instant text messaging could enhance both the processing speed and the comprehension of verbal contents [11]. Additionally, researchers have explored how facial features can serve as visual representations of multivariable data, allowing for nuanced interpretations of meaning. For instance, the technique known as Chernoff Faces encodes data points using facial features—such as the mouth, eyes, and eyebrows—so that viewers can detect anomalies through distinct facial characteristics [23]. Collectively, these findings underscore the potential of faces to serve as powerful, multidimensional tools for encoding complex attitudes.

As a potential channel for eliciting emotional attitudes, previous psychological research suggests that a face with an adjustable grin may capture attitudes more accurately by eliminating the need to translate subjective feelings into verbal responses [71]. However, this study also identified only a weak positive correlation between mouth curvature and subjective attitudes, suggesting that other facial features may also play significant roles in conveying emotions. This finding also aligns with research work by Boucher et al., who demonstrated that different facial regions contribute uniquely to the perception of specific emotions [10]. For instance, the brow and forehead are more strongly associated with sadness and surprise, while the cheeks and mouth are stronger indicators of happiness and disgust. More recent research has

further explored how humans perceive emotional signals in terms of both their category and intensity [20]. Specifically, the research revealed that facial movements used to classify different emotions are highly distinct, whereas those that intensify emotions tend to be more similar across categories. For instance, happiness is uniquely signaled by a smile, surprise by raised brows, fear by a stretched mouth, disgust and anger by an upper lip raise, and sadness by a lowered brow and raised chin. In contrast, intensifiers of emotions—such as mouth stretch—are shared across several emotions including surprise, fear, anger, and disgust, while cheek raising serves to intensify both happiness and sadness. These patterns suggest that emotional categories are differentiated by specific facial signals, whereas emotional intensity is conveyed through a common set of facial movements.

Building on these insights, future visual elicitation techniques could leverage the expressive potential of specific facial regions to more intuitively capture participants' attitudes. Instead of relying solely on inputs from Likert scales or sliders, such systems could allow users to manipulate an interactive facial interface—adjusting features like mouth curvature, the raising of brows, and cheek elevation—to visually express their emotional attitudes. At the same time, researchers could also apply the emotion classifiers and intensifiers discussed earlier to interpret both the type and intensity of the expressed attitude, enabling richer and analyzable representations of participant responses.

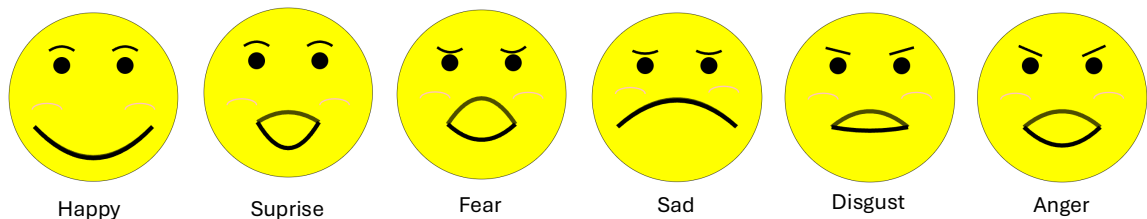


Figure 3.1: A face with adjustable eyebrows grin, and cheeks to elicit emotional attitudes (accessible in supplemental materials).

3.3.3.2 Incorporating verbal structure to elicit hedged attitudes

Text allowed participants to articulate the underlying rationale behind their hedged attitudes, frequently referencing personal experiences, sensory impressions, social norms, and other internal or external factors [99, 43]. Recent visualization research also highlights the importance of considering user preferences for text in information communication. Demographic differences, including variations in age, eyesight, and spatial skills, can impact visual information comprehension and shape the preferences for text vs. visuals [109, 50]. These preferences may also relate to how individuals form and express attitudes, which are often complex and layered [43]. Attitudes can be represented as networks of various evaluations on a given object [57]. Structured text elements, such as paragraphs and clauses, support the articulation of these layered perspectives by allowing individuals to logically develop and communicate nuanced arguments [62, 70], minimizing the cognitive load to map complex perspectives onto visual representations. Thus, while directional and emotional attitudes might be effectively represented through simple visuals such as facial expressions, conveying complex attitudes including hedging may require more sophisticated visual representations.

Building on this connection between structured text and attitude expression, future visual elicitation tools could explore ways to incorporate text-like structure into visual representations to better capture complex attitudes. As shown in Figure 3.2, we envision a set of interactive sliders as one such approach—designed to express both overarching attitudes and the caveats that contribute to hedging, in a manner analogous to how structured paragraphs support layered expression in text. This interface enables users to specify both positive and negative dimensions of a complex attitude. By editing the title of the main slider, users can express their primary attitude toward the topic. They can also add and then title additional sliders to represent sub-attitudes that serve as rationales supporting their main attitude. For instance, a

user could create a main slider labeled “Vaccination” to capture positive attitudes as the main stance related to disease prevention, alongside an additional slider labeled “Over-vaccinating Babies” to express potential concerns. This structure allows researchers to analyze the individual components of a hedged attitude—using the text labels and corresponding slider values—or to compute an overall attitude by averaging the slider values. Such an approach balances attitude expression with analytical clarity, enabling the collection of datasets that capture diverse sub-attitudes within each participant while remaining systematically comparable across different participants.

Vaccination

Main stance

Strongly Negative *Strongly Positive*

Over Vaccinating Babies *Strongly Negative* *Strongly Positive*

Sub-attitude Title *Strongly Negative* *Strongly Positive*

Add Sub-Attitude **Remove Topic**

Add Topic

Figure 3.2: A multi-slider technique to elicit hedged attitudes. (accessible in supplemental materials)

3.3.3.3 Using pictorial diagrams to elicit causal beliefs

Unlike verbal communication, pictorial language is universally effective [119]. Through our study, we observed that participants frequently used concrete pictorial items, which we refer to as pictorials in our code book, to depict attitudes and beliefs (60/82). This contrasts approaches taken in another popular form of visual communication - data visualizations, where designers prefer representing concrete objects in datasets using abstract encodings like shapes and lines [144]. This finding suggests that such abstract encodings may not naturally align with how people perceive their

surroundings. Instead, it suggests a deeper consideration for incorporating more concrete images, such as isotypes and pictographs [47, 16], in visual communication to better reflect people’s beliefs.

In our visual responses, pictorials are primarily used to convey complex attitudes and beliefs through vivid visual *stories* (28/82), which highlight the value of pictorials in visualizing complex mental constructs due to their flexibility and expressiveness. However, as researchers, eliciting attitudes/beliefs through visual stories can pose challenges, such as (i) difficulties incorporating them into more structured elicitation techniques and (ii) potential ambiguity in interpretation during subsequent analysis. To balance expressiveness and analyzability, we explored participants’ frequent use of diagrams as a more structured format for depicting causal relationships. Diagrams retain some expressive qualities of pictorials while also reducing the cognitive load associated with abstraction through drawing beliefs [121]. Given that beliefs often involve relationships between certain entities, diagrams might be particularly suitable for eliciting them, as they naturally support this relational structure, and enhance clarity in subsequent analysis [83].

Diagrams use literal and metaphorical conventions to convey meaning in multiple ways, with pictorials, spatial arrangements, and relationships between components all helping participants translate their mental constructs of beliefs into a visual format [29]. To maximize the effectiveness of utilizing diagrams, researchers should iteratively refine the diagrams, considering participants’ visual literacy and anticipating how they might interpret the visual representations within [29]. For instance, in our collected data, we observed that participants frequently used arrows to depict causal relationships between entities in a given topic (8/82). This usage aligns with the structure of argument maps, which support individuals in articulating complex reasoning by visually linking claims and subclaims through directional connectors such as arrows [118]. Another example is the use of directed acyclic graphs (DAGs),

which represent variables as nodes connected by arrows to illustrate assumed causal relationships [33]. Building on these insights, we envision a pictorial diagramming tool—illustrated in Figure 3.3—to further support the elicitation of beliefs as relational concepts. In this example, participants could use arrows to indicate whether an increase in one variable might lead to an increase or decrease in another. Unlike traditional argument maps or DAGs, this technique could be composed with others for richer elicitation techniques; e.g., as depicted in Figure 3.3, a slider could be used to adjust the strength of these relationships, visually showing the impact of one variable on another. As part of an iterative design process, further testing and refinement are needed to validate this tool’s effectiveness and generalizability across a broader range of users.

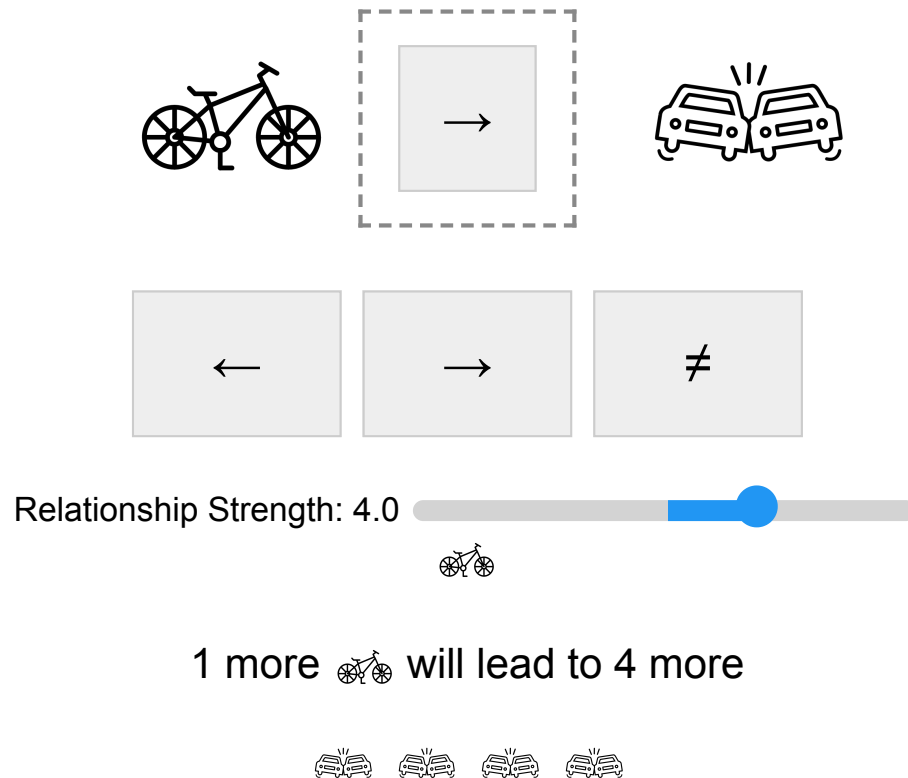


Figure 3.3: An interface to express the relationship between the use of bicycles and the occurrence of traffic accidents. (accessible in supplemental materials)

3.3.4 Eliciting Other Mental Constructs.

In this paper, we focused exclusively on exploring visual representations aligned with mental constructs of attitudes and beliefs. However, we hypothesize that our qualitative findings could also be valuable for other types of mental constructs. For example, previous research has used drawing to elicit *emotions* such as happiness in various scenarios, where participants often associate a happy face with visual depictions of leisure activities [79]. This aligns with our findings, where participants frequently use pictorials to depict their environment and faces to convey attitudes. Furthermore, structured diagrams such as tree maps, flow charts, and mind maps could be employed to elicit more structured abstract constructs, like *knowledge*, by offering a framework for organizing complex information [39]. Together, these insights underscore the potential of applying our findings to capture a wider array of mental constructs, highlighting further exploration within the visualization community.

3.4 Limitations and Future Work

Accuracy of Visual Elicitation Techniques. Our study revealed several visual approaches through which participants elected to represent their beliefs and attitudes. An important component of eliciting any mental construct is to enable people to accurately and faithfully represent their views. Our findings, including methods such as facial expressions, symbols, and diagrams, can inspire us to design new elicitation devices that can align well with how people think. An important limitation in our exploration, however, lies in determining *whether* and *to what extent* these different visual elicitation expressions can help people accurately represent their attitudes or beliefs. While our envisioned visual elicitation techniques have shown some potential to represent existing attitudes/beliefs, future research is needed to *evaluate* their effectiveness in eliciting attitudes and beliefs.

Not Everyone an Artist. In our study, 26 out of 41 participants reported they had no significant prior experience with drawing. Participants with more advanced drawing skills were able to create expressive *Stories* to communicate attitudes and beliefs. In contrast, participants with limited drawing abilities tended to produce oversimplified or ambiguous sketches, leading to difficulties in data interpretation. The skill level of participants in drawing influences the clarity of their visual responses and thus might introduce bias in our synthesized visual representations. To address this challenge in future research, standardized drawing templates or variety types of sketching examples could be provided to help all participants convey their attitudes/beliefs more effectively.

Differences in Visual Attitudes/Beliefs Across Cultures. Our study focuses on participants recruited exclusively from the United States, which may result in a relatively homogeneous cultural perspective rooted in Western values. This limitation is important because mental constructs such as beliefs and attitudes can vary significantly across cultural contexts. For example, while facial expressions such as smiles or frowns were commonly identified by our U.S. participants as effective means to convey attitudes, research has shown that the interpretation of these expressions differs across cultures [56]. Specifically, East Asian cultures may emphasize emotional intensity conveyed through the eyes, whereas Western cultures often prioritize the mouth when interpreting emotions. This cultural specificity suggests that the findings of this study may not be universally applicable and highlights the need for caution when generalizing across diverse populations.

Cultural differences also affect how individuals engage with elicitation tools. Studies on cross-cultural variations have found that East Asian participants are more likely to select the midpoint on scales, while North American participants tend to choose more extreme values [21]. This pattern has been linked to the Confucian principle of moderation, where East Asians may seek to avoid standing out and align more

with group norms, in contrast to the individualism often emphasized in North American cultures. Such differences in response tendencies could skew data distributions in cross-cultural research, leading to biased conclusions. Future studies should explore how to adapt visual elicitation tools to account for these cultural variations, potentially improving the accuracy and reliability of data collected from diverse populations.

Validation of the Elicitation Techniques. Evaluating the effectiveness of elicitation techniques is essential for both their design and selection [53]. However, the existing work in evaluating the effectiveness of leveraging visualization to elicit attitudes and beliefs remains limited [82]. One notable example is the work by Karduni et al., who assessed a visual elicitation technique designed to capture beliefs about linear correlations and their associated uncertainty [58]. Specifically, they compared the elicited belief distributions obtained through this technique with those generated using the Markov Chain Monte Carlo with People (MCMC-P) method [101], a widely used approach in cognitive science for eliciting subjective beliefs and estimating belief distributions. The evaluation strategy employed by Karduni et al. offers a valuable model for future work, particularly in assessing the **analyzability** of visual elicitation techniques by benchmarking visually elicited data against distributions derived from structured, iterative methods such as MCMC-P.

Equally important is evaluating the **expressiveness** of visual elicitation techniques: the degree to which they allow participants to communicate nuanced, multifaceted perspectives. Researchers could utilize the Elicitation Interview technique to gather and understand participants' experiences, such as their emotional reactions to the topic and the personal meaning it holds when expressing attitudes and beliefs through visual elicitation techniques [54]. By explicitly asking participants to what extent the visual elicitation method helped them express these rich insights, researchers can gain a deeper understanding of the technique's expressive capacity.

3.5 Summary

In this chapter, we presented findings from a two-round qualitative study involving 41 participants. This study uncovers the nature of participants’ mental constructs of attitudes and beliefs and the methods with which they intuitively externalized their attitudes and beliefs, encompassing both visual and verbal channels. We identified key elements that serve as the building blocks of participants’ verbal and visual responses, and distilled a spectrum of strategies they used, providing insights into both the ‘what’ and the ‘how’ of communicating attitudes and beliefs. Based on these findings, we envisioned alternative visual elicitation prototypes, which could inform the design of future survey instruments. By offering a broader range of expressive options—both verbally and visually—researchers and designers can create more intuitive, engaging, and effective interfaces for capturing and understanding user attitudes and beliefs.

Chapter 4

Evaluating the Impact of Textual Summary and Annotation on Confirmation Bias in Data Visualization Interpretation

Textual summaries of visual patterns and accompanying annotations can potentially impact how users interpret visualizations, depending on whether or not they align with users' prior beliefs. This chapter presents work conducted in response to **Research Question 2**, which investigates how these two key design choices in static visualizations contribute to confirmation bias. This work was accepted as a full paper at ACM CHI 2025 [75].

RQ 2: How do textual summaries and visual annotations influence confirmation bias in interpreting static visualizations?

Textual annotations—text-based notes that explain or comment on visual patterns—can help readers efficiently detect relevant data features [110, 6, 59] and can even shape how the data are interpreted [111, 66, 65]. Among various forms of textual

annotations, we focus on data facts—concise textual summaries that highlight insights ranging from single values to complex comparisons of distributions—because they closely align with specific data trends. These summaries often resemble the kinds of answers users seek during low-level visual analysis tasks, such as identifying extrema, trends, or comparisons [108, 1], making them especially influential in shaping how people interpret data. In many cases, textual annotations in visualizations are also accompanied by visual annotations—graphical elements that emphasize specific data points or patterns—to further direct attention and improve clarity [95, 104, 19, 96]. In this work, we focus primarily on how data facts influence confirmation bias in visualization interpretation, while also examining how the presence of visual annotations may amplify or modulate that effect.

To answer **Research Question 2**, we examine three key aspects of data facts: their *presence*, *presentation style*, and *strength*. Through a series of three preliminary studies and one primary experiment, we investigate how confirmation bias may be exacerbated or mitigated by data facts (in our experiment design, we operationally refer to data facts as brief textual summaries highlighting statistical patterns in visualizations). To better understand the factors influencing confirmation bias in visual data communication, we draw on a range of psychological and data visualization research to formulate hypotheses.

First, we investigate the effect of **prior belief** on confirmation bias. Previous research in psychology has shown that confirmation bias can fluctuate in severity depending on the relationship between the user’s stance and the displayed information. For example, confirmation bias is more pronounced when people hold stronger attitudes [15]. Hence, we examine whether the impact of prior belief extends to the interpretation of visual communication. This will also help us account for the effect of prior belief in later analyses to reach more generalizable conclusions on the effect of other experimental factors.

H1 - Prior Belief Strength: Participants with *stronger* prior beliefs will exhibit *stronger* confirmation bias.

Similar to prior beliefs, the data topic is another contextual factor that is difficult to control when visualization researchers build tools or generate best-practice guidelines for visual analysis in the real world. Data topics might interact with the strength of one’s prior beliefs to consequently impact the degree of confirmation bias. For instance, political or polarizing topics in which people tend to have stronger prior beliefs may be more likely to exacerbate confirmation bias than more neutral topics [122, 32, 135].

H2 - Topic: Confirmation bias will be *higher* with a more *polarizing* dataset (COVID-19 Vaccination) compared to a less polarizing dataset (Diet Choices).

With the previous two hypotheses and subsequent analysis, we can build models to control for the effect of prior belief and data topics to investigate the role of data facts in influencing confirmation bias. Data facts can either *support* or *refute* one’s prior beliefs to differently affect confirmation bias. Prior research has revealed that individuals tend to perceive a visualization as less credible when the title is *inconsistent* with their prior beliefs compared to when the title is *consistent* with their beliefs [65]. We thus test how **data facts and belief alignment** impacts the manifestation of confirmation bias.

H3 - Data Facts Alignment: Confirmation bias will be higher when visualizations are accompanied by data facts that *support* participants’ prior beliefs, and lower when the visualizations are accompanied by data facts that *refute* participants’ prior beliefs, compared to when the visualizations are shown without any data facts.

We also examine the presentation style of data facts. Increasingly, visualizations in analytic environments are accompanied by automatically generated data facts to aid interpretation [108, 131]. Recent advancements have even enabled the automatic generation of visual annotations based on textual descriptions, streamlining the an-

notation process [72]. These developments motivate our investigation into how the presence of visual annotations—used to visually reinforce the patterns described by data facts—may influence confirmation bias in data interpretation.

H4 - Data Facts Presentation Style: Confirmation bias will be strongest for participants who view data visualization with *supporting* textual data facts only, followed by textual data facts with visual annotations, and then *No Data Facts*. For *refuting* data facts, confirmation bias will be lower when participants view visualizations with textual data facts only, followed by textual data facts with visual annotations.

Finally, we consider the **strength** of the data facts themselves. Previous studies have demonstrated that captions highlighting *key visual features* in data visualizations make these features more salient [138]. Readers often interpret these captions as the primary chart takeaway [59]. Building on this, we examine whether this effect extends to data facts. We generated a set of data facts that describe different insights, referencing the taxonomies proposed by Amar et al. [1], to create a ranked list of data facts varying in persuasive **strength**. We test the effect of data facts on confirmation bias based on their strength.

H5 - Data Facts Strength: Confirmation bias will be higher when ambiguous visualizations are presented with *supporting stronger* data facts (i.e., describing correlations) compared to *supporting weaker* data facts (i.e., describing a derived value). Confirmation bias will be lower when visualizations are presented with *refuting* correlation data facts compared to *refuting* derived value data facts.

Summary of Findings. A summary of our findings can be found in Table 4.1. Our results showed that **data facts can exacerbate confirmation bias when their phrasing supports an individual’s prior beliefs, and this effect is further exacerbated when accompanied by visual annotations.** Conversely, data facts can *mitigate* confirmation bias when their phrasing *refutes* an individual’s prior be-

Hypothesis		Support	Finding Summary
H1	Participants with <i>stronger</i> prior beliefs will exhibit <i>stronger</i> confirmation bias.	Supported	The strength of pre-existing beliefs is positively correlated with the level of exhibited confirmation bias.
H2	Confirmation bias will be <i>higher</i> with the more <i>polarizing</i> dataset (COVID-19 Vaccination) compared to the less polarizing dataset (Diet Choices).	Not Supported	The polarization level of the visualized topic does not have an effect on confirmation bias.
H3	Confirmation bias will be higher when visualizations are accompanied by data facts that <i>support</i> participants' prior beliefs, and lower when the visualizations are accompanied by data facts that <i>refute</i> participants' prior beliefs, compared to when the visualizations are shown without any data facts.	Supported	<i>Supporting</i> Data Facts lead to a higher level of confirmation bias compared to when there are <i>No Data Facts</i> . <i>Refuting</i> Data Facts result in a lower level of confirmation bias compared to when there are <i>No Data Facts</i> .
H4	Confirmation bias will be strongest for participants who view data visualization with <i>supporting</i> textual data facts only, followed by textual data facts with visual annotations, and then <i>No Data Facts</i> . For <i>refuting</i> data facts, confirmation bias will be lower when participants view visualizations with textual data facts only, followed by textual data facts with visual annotations.	Partially Supported	<i>Supporting</i> Data Facts with Visual Annotation lead to a greater level of confirmation bias compared to when there are <i>No Data Facts</i> . No significant difference was identified between <i>Refuting</i> Data Facts with Visual Annotation, <i>Refuting</i> Data Facts only, and <i>No Data Facts</i> .
H5	Confirmation bias will be higher when ambiguous visualizations are presented with <i>supporting</i> strong data facts (i.e., describing correlations) compared to <i>supporting</i> less strong data facts (i.e., describing a derived value). Confirmation bias will be lower when visualizations are presented with <i>refuting</i> correlation data facts compared to <i>refuting</i> derived value data facts.	Partially Supported	<i>Supporting</i> Correlation Data Facts lead to a higher level of confirmation bias compared to <i>No Data Facts</i> . <i>Refuting</i> Correlation Data Facts result in a lower level of confirmation bias compared to <i>No Data Facts</i> .

Table 4.1: Results on Research Question Hypotheses. Data facts that *support* prior beliefs tend to intensify confirmation bias, while those that *refute* prior beliefs can mitigate it. Data facts describing key visual features can further amplify this effect. Additionally, visual annotations with *supporting* data facts exacerbate confirmation bias. Rows highlighted in blue indicate hypotheses that received partial or full support.

liefs. Our findings highlight a critical drawback of incorporating textual summaries into data visualizations: data facts can introduce bias, particularly when they align with individuals' prior beliefs. However, we also found that data facts show promise in bias mitigation when the data fact supports a refuting hypothesis. We discuss the implications of these findings for visualization design and suggest that careful consideration is needed when adding textual annotations—whether as titles or descriptions—to ensure that the interpretation remains balanced and not overly influenced by these textual elements.

4.1 Study Overview

In this work, we examine the effect of visual and textual annotation on confirmation bias in visualizations. This type of investigation is sensitive to contextual factors

such as the visualization topic, the complexity of the data visualization, the patterns in a visualization that the data facts describe, and our participants' prior beliefs. Therefore we conducted a series of preliminary studies (PS) to identify the optimal experimental parameters for our main study, including:

- *PS 1*: a range of topics with various prior belief distributions among participants,
- *PS 2*: visualization stimuli that are *ambiguous* and, according to an individual's prior beliefs, could be selectively interpreted as supporting or refuting, and
- *PS 3*: a set of data facts with different strengths to explore their impact on confirmation bias.

We conducted our main study using experimental parameters determined by the preliminary study. We provide a list of the dependent variable (DV) and independent variables (IV) we examined for the impact on confirmation bias in our main study.

- *DV*: amount of confirmation bias
- *IV 1 (H1)*: prior belief strength
- *IV 2 (H2)*: chart topics (vaccination, dietary choices)
- *IV 3 (H3)*: data facts alignment (supporting, refuting)
- *IV 4 (H4)*: data facts presentation style (none, textual, textual + visual)
- *IV 5 (H5)*: data facts strength (correlation, derived value)

4.2 Preliminary Studies

Participants for all preliminary studies were recruited through the crowdsourcing platform Prolific [90]. We screened for participants who are based in the United States

who are fluent in English and have a study approval rate greater than 95%. Individuals who participated in any one study were ineligible to participate in subsequent ones. The study plans were reviewed and approved by the authors’ university ethics board.

4.2.1 Preliminary Study 1: Data Topic Selection

Existing beliefs significantly impact how one processes information. The topic of a visualization can trigger varying beliefs, making the reader more or less susceptible to confirmation bias [51]. Topics unfamiliar to readers are less likely to elicit confirmation bias due to a lack of pre-existing beliefs, whereas polarizing or political topics are more likely to elicit confirmation bias as individuals’ entrenched stances are less likely to move [83]. To enhance the robustness of our work, we tested various topics to identify ones associated with strong and weak prior beliefs. The main study uses topics selected from this set. This approach allows us to account for the effects of prior belief strength.

Participants. We recruited 40 participants from the crowdsourcing platform Prolific ($M_{age} = 33.59$, $SD_{age} = 12.47$; 20 Male, 18 Female, 1 Non-binary, 1 Prefer not to disclose).

Tasks & Procedure. Participants were presented statements on 16 topics (e.g., vaccine, dietary preference, government surveillance) in a random order, such as “COVID vaccination should be a mandatory policy.” The full list can be found in Figure 4.1. Following a similar set-up as prior work that used belief elicitation (e.g., [140]), we asked participants to rate the “extent to which they agree with the statement” on a scale ranging from “-100 = Strongly Disagree” to “0 = Neutral” to “100 = Strongly Agree.” The study ended with several optional demographic questions including their age and gender.

Measuring Topic Polarization. The data we collected presents a distribution of scores for each topic. We employed a continuous slider labeled with "Strongly Disagree" on the left, "Neutral" in the center, and "Strongly Agree" on the right. The numerical range of the slider was not visible to participants, who only saw the labels on the slider ("Strongly Disagree", "Neutral", and "Strongly Agree"). We categorized scores below 0 into the 'Disagree' cluster, and those exceeding 0 in the 'Agree' cluster. We aimed to select a topic associated with strong, polarizing beliefs (so that regardless of participants' stances the topic would elicit strong beliefs), as well as a neutral topic associated with weak beliefs. To this end, we computed the Cluster-Polarization Coefficient (CPC) [85] to measure the polarization of each topic. CPC captures two important features of political and social polarization: intergroup heterogeneity (BSS , which represents the variance between each cluster) and intragroup homogeneity (WSS , which represents the sum of variance within all clusters). Subsequently, we employed Equation 4.1 to compute the CPC score. The score indicates the polarization level for each topic, with 0 denoting no polarization (e.g., all participants consistently adjusted the slider toward either the left (disagree) or right (agree) side of the central point "Neutral," indicating unanimous disagreement or agreement with the statement) and 1 indicating extreme polarization (e.g., all participants in the disagree cluster moved the slider to the exact same position at the left side of the central point, while all participants in the agree cluster moved it to the exact same position at the right side of the central point). Responses with neutral beliefs were excluded when calculating CPC (3.4% of total responses), as our goal was to explore polarization on both the 'Disagree' and 'Agree' sides of each topic.

$$CPC = 1 - \frac{WSS}{WSS + BSS} = \frac{BSS}{WSS + BSS} \quad (4.1)$$

Results. We ranked all topics based on the computed CPC scores (full details

in supplementary materials). One potential limitation of the CPC metric in our experiment setting could be that a CPC of 1 indicates extreme polarization, but does not necessarily reflect strongly held beliefs, as it could also result from participants agreeing and disagreeing uniformly on weak points. Consequently, we validated our topic selection by examining the belief distributions (as shown in Figure 4.1) to ensure the selected topics represent both the desired polarization *and* strength of belief. The topics we ended up selecting include:

- High polarization topic associated with stronger beliefs: *COVID-19 Vaccination* (CPC: 0.74).

We selected ‘Vaccination’ as the polarizing topic based on its highest CPC score and the concentration of strong beliefs within both the agree and disagree clusters.

- Neutral topic associated with weaker beliefs: *Diet Choices* (CPC: 0.65).

We selected ‘Diet’ as the neutral topic because it has the highest percentage of beliefs near the neutral point (0), with 15% of participants within the [-5, 5] range. Although two other topics, ‘Immigration’ (CPC = 0.61) and ‘Work from Home’ (CPC = 0.51), also have 15% of beliefs near neutral, ‘Diet’ maintains a higher level of polarization (0.65). We chose the topic with the higher CPC score because it reduces the likelihood of a skewed belief distribution (e.g., topic 16-Pet in Figure 4.1) among participants, which could hinder their reception to evidence that contradicts participants’ common assumptions.

4.2.2 Preliminary Study 2: Visualization Stimuli Selection

Existing work has operationalized confirmation bias as the tendency to interpret ambiguous evidence in a way that supports one’s prior beliefs [63, 18, 28]. Building on this idea, we measure confirmation bias by examining how much participants perceive

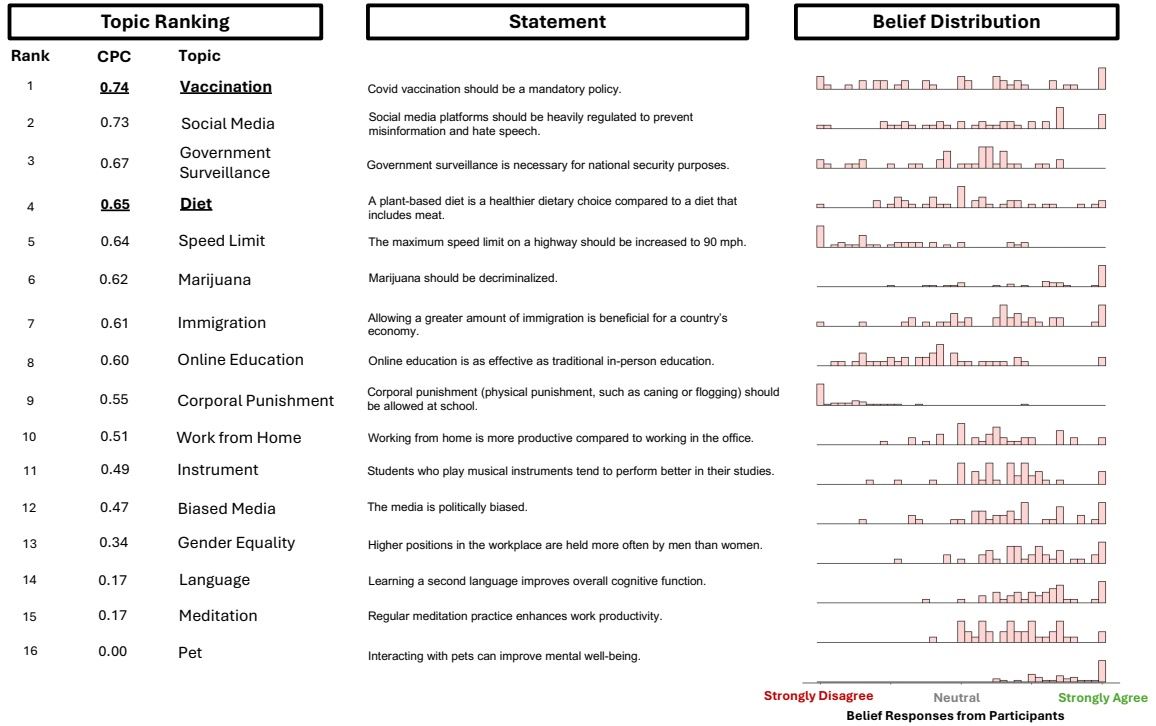


Figure 4.1: Preliminary Study 1 Results and Belief Distributions. Participant responses were collected as slider inputs and visualized as histograms for each topic, with a bin width of 5 for each bar. The x-axis ranges from -100 to 100, with -100 representing ‘Strongly Disagree,’ 0 representing ‘Neutral,’ and 100 representing ‘Strongly Agree.’ We selected ‘Vaccination’ as the polarizing topic, as individuals tend to hold strong prior beliefs about it, and ‘Diet’ as the neutral topic, where people generally hold weaker beliefs.

an ambiguous data visualization—one that neither strongly supports nor refutes an existing belief—as supporting their own beliefs. A visualization can contain multiple patterns. Readers can draw different inferences about data depending on what patterns they pay attention to [6]. To conduct our main study, we need a set of ambiguous visual stimuli that are just complex enough to allow for various interpretations but still straightforward for participants to engage with. This preliminary study tackles this goal via two stages. First, we identify the optimal data set size so the visualization is complex but not overwhelming. Second, we tweak the data values to generate ambiguous patterns in data that can be interpreted in competing directions. Our goal is to create stimuli with evenly divided interpretations: approximately 50% of the responses should indicate seeing a predominantly increasing trend, and the rest a decreasing trend. Using ambiguous visualizations for our study not only provide us the max amount of space to observe variance in data patterns to avoid floor and ceiling effects, but also provides a clean, conservative measure of confirmation bias as the observed confirmation bias will be less influenced by the visualization’s inherent narrative. If we instead used a non-ambiguous visualization that explicitly takes a stance, we would need to model its relative position to each participant’s belief, introducing additional complexity and noise.

4.2.2.1 Stage 1: Determining Small Multiples Size.

Materials: Prior work suggested that bar charts depicting groups of data can elicit ambiguous takeaways [111]. Therefore we generated four types of faceted bar charts: arranged in either 3×3 or 5×5 grids, with each facet cell within these grids featuring either a single bar or two bars representing different groups. To introduce ambiguity, we balanced the number of rows displaying strictly ascending and descending trends in bar heights. We also incorporated rows that presented neutral trends (an equal

number of descending and ascending trends between adjacent cells). For each design, we created two datasets to further examine whether the ambiguity persisted across different data sets. All stimuli from this stage are included in the supplemental materials.

Participants: We recruited 83 participants ($M_{age} = 39.23$, $SD_{age} = 13.84$; 44 Male, 39 Female) from Prolific. Each participant was randomly assigned to one of our visualization stimuli (3x3 single-bar, 3x3 multi-bar, 5x5 single-bar, or 5x5 multi-bar).

Study Procedure: We designed a visual search task to evaluate the complexity of each visualization following guidelines outlined in Elliott et al. [38]. Each participant went through three visual search tasks in a random order. Each task involved presenting a visualization and asking users to determine whether it displayed an increasing or decreasing trend in row-wise bar heights using a two-choice forced response. We counterbalanced the order of the response options (descending first or ascending first) to prevent an ordering effect. Two of the tasks used the faceted bar charts we designed. The third task served as a comprehension check, using a visualization that depicted a strict descending trend across all rows, to assess the participants' ability to extract trends from visualizations. To avoid the impact of participants' pre-existing beliefs on data interpretation, we created hypothetical scenarios about alien fictional gases (Gas Task) and chemicals in lake (Chemical Task) for this study. We provide one example task below. The rest can be found in the supplementary materials.

- *Example Task (Gases Task):* You need to determine which of two fictional gases, Zytharane or Skylox, is dominant in a sample consisting exclusively of these gases. Zytharane is characterized by a warming effect, where its concentration increases with rising temperatures. Conversely, Skylox exhibits a cooling effect, with its concentration decreasing as temperatures rise. This data visualization depicts the relationship between the atmospheric concentration of the gas mixture and temperature changes. To examine the dominant gas in

mixture S, please analyze whether the visualization shows an overall ascending or descending trend in the atmospheric concentration as the temperature increases.

Each participant was forced to wait a minimum of 20 seconds before answering to encourage a proper examination of each visualization. After completing the three tasks, they were asked demographic questions including age, gender, and the highest level of education.

Results: As shown in Figure 4.2 A , the 5×5 multi-bar condition led to the most even split in participant response. We took a closer look at participant feedback from Prolific direct messages. Many indicated that the multi-bar visualization was confusing, and as a result, they incorrectly approached the task by comparing the two bars within the same cell for ascending or descending trends, rather than comparing the overall bar heights across the same row. Further, the 3×3 seemed was too simplistic of a visualization, such that participants overall saw either only the ascending or the descending trend (depending on the dataset). Therefore, we decided to go with the 5×5 single-bar visualization.

4.2.2.2 Stage 2: Generating Ambiguous Data

In order to further improve our 5×5 single-bar visualization to be consistently ambiguous, we adjusted the specific data values to reach an even 50-50 split amongst participants in terms of reported ascending or descending trends.

Materials. We developed a Python script that simulates data values in a 5×5 matrix to test a range of stimuli. The script enforces the balance of the following patterns for each row to maintain ambiguity (see supplemental materials for more details):

- **Pattern 1 Strictly Ascending/Descending:** The heights of the five bars in a row progressively increase or decrease from left to right.

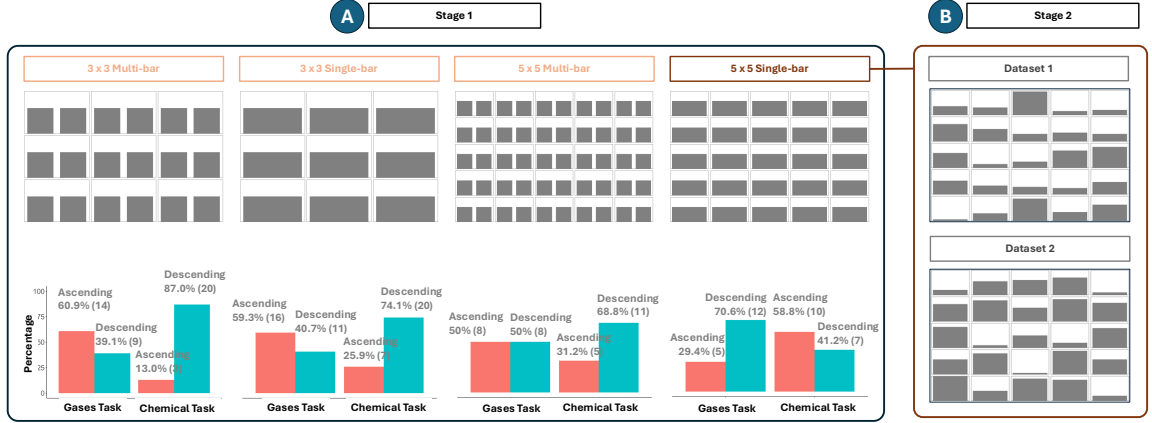


Figure 4.2: Preliminary Study 2: Stage 1 (A) and Stage 2 (B) Conditions. In Figure 4.2 A, the depicted charts under assessed dimensions are simplified graphical representations of the stimuli, the actual depicted charts in our study include titles, axis labels, and tick marks (see supplemental materials). At the bottom of Figure 4.2 A, the bar charts illustrate the distribution of evaluations for each stimulus dimension in Stage 1. We selected 5×5 single-bar stimuli, colored in brown, and chose datasets 1 and 2 from the 16 generated datasets in Stage 2 as depicted in In Figure 4.2 B.

- **Pattern 2 Overall Ascending/Descending:** For a given row, four out of five bars follow a strict ascending or descending sequence from left to right.
- **Pattern 3 Neutral trend:** For a given row, the height difference between any symmetrical pair of bars (e.g., the first bar on the left and the last bar on the right, the second and the fourth bars) will not exceed 3 units. This specification keeps enough variability in the neutral trend to appear realistic.

To ensure the ambiguity of each resulting visualization, the script ensures there exists an equal number of rows following Patterns 1 and 2. We simulated 16 ambiguous data visualizations that satisfy our criterion of ambiguity.

Participants. We recruited 100 participants ($M_{age} = 37.92$, $SD_{age} = 12.38$, 45 Male, 53 Female, 1 Non-binary, 1 Prefer not to disclose) from Prolific and conducted two rounds of this preliminary study (50 participants, 8 visualizations in each round).

Study Procedure. In each round, participants examined eight ambiguous visualizations in random order and answered the question, "Based on the visualization

provided, please identify the general trend in bar heights as you move from left to right across each row.” Participants then selected one of two options: Ascending or Descending. To ensure a thorough evaluation, each participant was required to wait a minimum of 20 seconds before responding. We also randomly inserted a control visualization with five strictly ascending rows as a comprehension check to assess each participant’s ability to reason whether the majority of rows depict an overall ascending or descending trend. Participants who failed the comprehension check were excluded from the data analysis.

Results. An ideal ambiguous visualization would be one in which participants responded in equal proportions that the visualization depicted an ascending v. descending trend (i.e., 50% ascending, 50% descending). To account for noise, we consider stimuli sufficiently ambiguous if participants’ response distributions fell between 40% and 60% ascending/descending. Five stimuli satisfied this condition. Among them, we selected the two visualizations that elicited responses closest to 50% ascending/descending, as depicted in Figure 4.2 B.

4.2.3 Preliminary Study 3: Strength of Data Facts

Data facts can vary in their persuasive strength [111, 91]. To account for the effect of data fact strength on confirmation bias, we conducted this study to identify data facts with strong and weak persuasiveness.

Participants. We recruited 40 participants ($M_{age} = 33.35$, $SD_{age} = 9.29$; 21 Male, 17 Female, 1 Non-binary, 1 Prefer not to disclose) from Prolific. Each participant read visualizations with ten types of data facts in random order.

Materials.

Data Facts: We designed ten types of data facts aligned with the ten low-level analytic tasks delineated in Amar et al.’s framework, similar to the approaches by Srinivasan

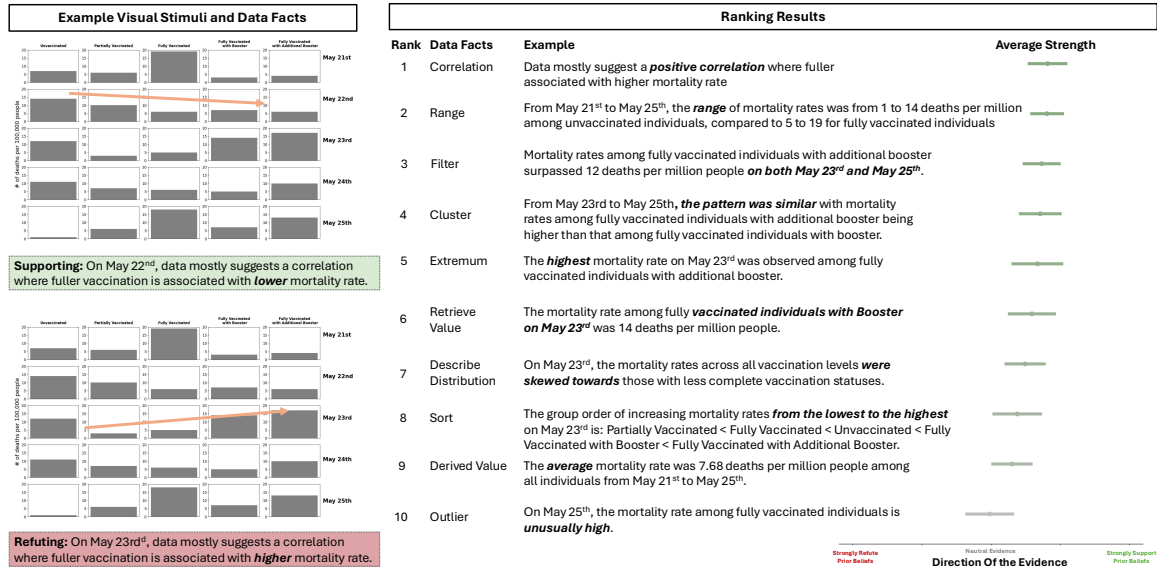


Figure 4.3: Preliminary Study 3: Data Facts and Results. The strength of each type of data fact was assessed based on how strongly participants perceived the data visualizations and facts as supporting their prior beliefs. The mean value and 95% confidence intervals for each type of data fact are depicted in the confidence interval plot. The color coding indicates whether the mean rating of each data fact leans toward supporting or refuting participants' prior beliefs.

et al. [108, 1]. These examples are included in the supplemental materials.

Visualization: We used one of the 5×5 small multiples bar charts as depicted in Figure 4.2, depicting the vaccination topic. This visual stimulus is also included in the "Example Visual Stimuli and Data Facts" panel in Figure 4.3.

Study Procedure. After obtaining informed consent, each participant read 10 visualizations, each accompanied by a line of text describing a unique data fact, in random order. We employed the same slider settings as described in Section 4.2.1 to elicit participants' prior beliefs on vaccinations with the prompt: "Please rate the extent to which you agree with the following statement: COVID vaccination should be a mandatory policy." To measure the *confirmatory* effect of each type of data fact, we manipulated all data facts to align with participants' prior beliefs. We also included an attention check question: "If you are paying attention, please drag the handle to the left end of the slider. This is an attention check." For each

visualization, participants evaluate it by responding to the prompt: *"How strongly do you perceive this visualization and the accompanying text to **support** the takeaway: COVID vaccination should be a mandatory policy?"* Participants responded using an interactive slider ranging from strongly refuting (-100) to strongly supporting (100). The numerical ranges for both sliders, eliciting prior beliefs and visualization evaluations, were hidden from participants.

We also included a comprehension check at the beginning of the study to ensure participants understood how to read our visualization stimuli. We asked: *"What was the mortality rate on May 22nd among partially vaccinated individuals?"* as a multiple-choice question. Participants who failed the attention check or the comprehension check were excluded from the analysis. Participants who held a neutral belief were also excluded as this makes it impossible to interpret potentially confirmatory belief updating.

Results. For each data fact, we calculated its perceived data fact strength score following Equation 4.2. The term *Response* represents the input from each participant's interactive slider. The data fact strength score reflects the extent to which the participant perceived the visualization (which was designed to be ambiguous and neutral) to be aligned with their prior beliefs. We designed the experiment such that participants only interacted with visualizations containing supportive data facts for evaluating confirmatory effects. A positive data fact strength score (+) means the participant reported the visualization to align with their prior beliefs, suggesting that the data fact biased the visualization toward those beliefs. Conversely, a negative score (-) means the participant reported the visualization to refute their beliefs, indicating that the data fact biased the visualization against those beliefs despite the data fact being designed to be supportive.

$$Data\ Facts\ Strength = \pm |response| \quad (4.2)$$

We calculated the average strength score for each type of data fact across all participants to produce a ranking of the strength for each data fact, as summarized in Figure 4.3. We ultimately selected ‘Correlation’ as the strong data fact and ‘Derived

Table 4.2: The statement for each topic

Topic	Statement
COVID-19 Vaccination	COVID vaccination should be a mandatory policy.
Diet Choices	A plant-based diet is a healthier dietary choice compared to a diet that includes meat.

Value’ as the weak data fact for our main study. We used ‘Derived Value’ instead of ‘Outlier’ as the data facts with the lowest strength because the ‘Outlier’ condition resulted in a negative score, which could suggest unfamiliarity with the concept and potential misunderstanding. Hence we avoided using this data fact type.

4.3 Main Study

After the three preliminary studies, we conducted our main study to address our research questions in a pre-registered experiment¹. Figure 4.4 shows the experimental variables and procedure. A summary of our findings is presented in Table 4.1, with more detailed analyses available in the supplemental materials.

Materials.

- *Topics:* To validate the impact of data facts across a broader range of pre-existing belief strengths, we selected two topics as determined from the Preliminary Study 1:
 - **[COVID-19 Vaccination]** A more polarizing topic where individuals tend to hold strong beliefs either in support of or opposition to mandatory COVID-19 vaccination.

¹https://aspredicted.org/N78_C7G

- **[Diet Choice]** A less polarizing topic where individuals tend to have relatively more neutral beliefs on whether a plant-based diet is healthier than one that includes meat.
- *Visual Stimuli:* Based on the results of Preliminary Study 2, we selected the ambiguous 5×5 small multiples bar charts with a single bar per cell as the stimuli, with two synthetic datasets.
- *Data Facts:* Two types of data facts with varying strengths accompanies our visualizations, based on the results of Preliminary Study 3.
 - **[Correlation]** A more compelling data fact describing the correlation between column and row variables.
 - **[Derived Value]** A less compelling data fact describing the average value of certain columns.

Conditions. Participants were randomly assigned to one of three presentation styles as between-subjects conditions: textual data facts with visual annotations, textual data facts only, or *No Data Facts*. In all conditions, each participant read two visualizations, with the topic and datasets counterbalanced with the display order, as shown in Figure 4.4. For those in the textual data facts only and the textual data facts with visual annotations conditions, each participant read both a visualization with a correlation data fact and one with a derived value data fact, one of which supports their prior belief and the other refuting. Their display order and combination are counterbalanced. For example, if the first visualization showed a supporting correlation data fact on the topic of vaccines via dataset 1, the second visualization showed a refuting derived value data fact on the topic of diet choices via dataset 2. For the *No Data Facts* condition, participants viewed the two ambiguous visualizations without any data facts, with the dataset topic order randomized.

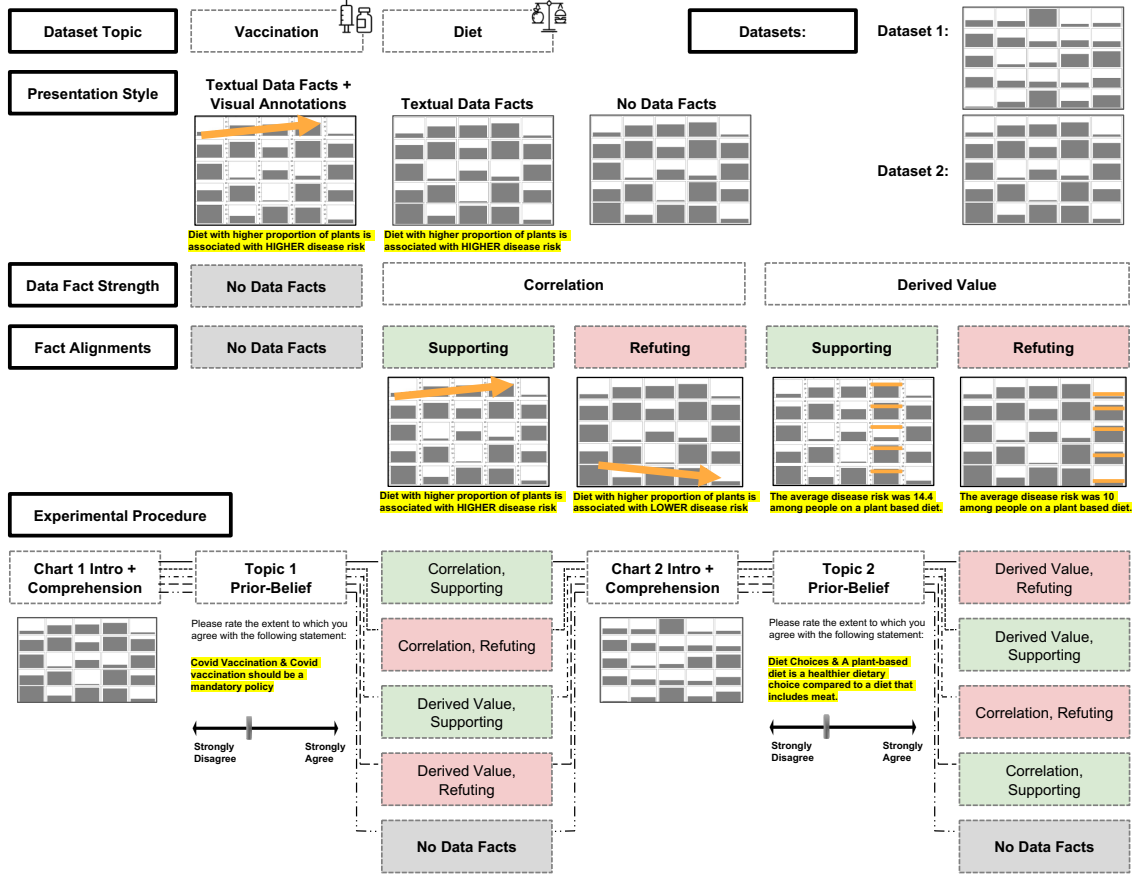


Figure 4.4: Main Study Conditions and Procedure: [Presentation Style] is a between-subjects condition. Each participant completes 2 trials, either involving 2 data visualizations *with data facts*, where [Dataset, Dataset Topic, Data Facts Strength, Fact Alignment] are within-subject conditions, or 2 visualizations with *No Data Facts*, with [Dataset Topic] as the only within-subject condition. The varying types of strokes connecting the components in the main study procedure diagram represent the different conditions assigned to each participant.

Task and Procedure. Our study procedure is shown in Figure 4.4. After completing the consent form, participants start with a warm-up activity to familiarize them with our stimuli. During the warm-up, they read chart descriptions explaining information encoded by rows and columns, followed by a comprehension check to ensure they understood how to interpret the chart. Participants who do not pass this check are excluded from our analysis.

Next, we assessed participants' pre-existing beliefs about each topic using an interactive slider ranging from "strongly disagree" (-100) to "strongly agree" (100) with

the prompt: "Please rate the extent to which you agree with the following statement:" Statements related to the topics were provided as outlined in Table 4.2. Participants then read the bar charts, either with a textual data fact, with a textual data fact plus visual annotations, or with *No Data Facts*. Afterward, they were prompted to rate their level of agreement with the same statements in Table 4.2, using an interactive slider ranging from "strongly *refute*" (-100) to "strongly *support*" (100). The numerical ranges of both sliders were not visible to participants. An additional trial on the topic "Productivity while working from home" was included as an attention check and appeared randomly between the two main trials: *"If you are paying attention, please drag the handle to the left end of the slider. This is an attention check."* Participants who failed this check were excluded from our analysis. Participants who held a neutral belief were also excluded as this makes it impossible to interpret potential confirmation bias.

Measuring the Strength of Prior Beliefs: We measure the strength of each participant's prior belief on each topic as the absolute distance from either endpoint of the slider ("strongly disagree" and "strongly agree"), based on the following equation:

$$belief\ strength = |initial\ belief| \quad (4.3)$$

In Equation 4.3, *initial belief* represents the participant's response from the interactive slider. If a participant *agrees* with the statement provided, we measure the strength as the absolute distance from the right end of the slider. Conversely, if a participant *disagrees* with the statement, we measure the strength as the absolute distance from the left end of the slider.

Measuring Confirmation Bias. We evaluated how participants interpreted an ambiguous data visualization in relation to their prior beliefs on each topic to assess confirmation bias. On the slider, a value of 0 indicates a neutral stance towards the

statement in the prompt. Therefore, the confirmation bias score was calculated using the formula:

$$bias\ score = \pm |evaluation| \quad (4.4)$$

In this equation, *evaluation* is the participant’s rating of how much the ambiguous data visualization either supports or refutes the statement given in the prompt. The bias score is assigned a positive (+) sign when the evaluation of the visualization aligns with the participant’s prior beliefs and a negative (-) sign when it contradicts those beliefs.

Participants. We recruited 1,080 participants from Prolific ($M_{age} = 38.79$, $SD_{age} = 12.96$; 531 Male, 537 Female, 5 Non-binary, 2 Transgender, 1 Genderqueer, 1 Genderfluid, 3 Prefer not to disclose). We screened for participants who are based in the United States who are fluent in English with study approval rate greater than 95%. Individuals who participated in preliminary studies were ineligible to participate in the main study. The study plan was reviewed and approved by the authors’ university ethics board. The sample size was determined by a power analysis conducted using G*Power [41] to ensure a minimum power of 0.75 (indicating a 75% chance of detecting a true effect for each hypothesis if it exists) across all effect sizes from our hypotheses, with an alpha level of 0.05. This analysis was based on a pilot study involving 61 participants under the same settings as the main study.

4.4 Results

See Table 4.1 for a summary of results.

Methodology. We use a single mixed-effects linear model to test H1, H2, and H3. For H4 and H5, we separately examine the effects of supporting and refuting data facts using two linear models:



Figure 4.5: Main Study Hypotheses and Results. The results of **H1** are illustrated as a scatterplot showing the relationship between the strength of prior beliefs and confirmation bias. The *estimated marginal means* from each linear model and the 95% confidence intervals of confirmation bias scores across different conditions in **H2** - **H5** are plotted as confidence interval plots. For **H2**, colors represent different topics, while for **H3**, **H4**, and **H5**, dark red indicates *refuting* data facts, forest green represents *supporting* data facts, and grey signifies no data facts.

- **Situation SN:** Analyzing data that includes only participants who viewed visualizations with *Supporting* Data Facts and *No Data Facts*.
- **Situation RN:** Analyzing data that includes only participants who viewed visualizations with *Refuting* Data Facts and *No Data Facts*.

This separation is crucial, as we hypothesize that *supporting* data facts increase confirmation bias compared to the no data fact condition (our baseline), raising the score above zero. On the other hand, *refuting* data facts decrease confirmation bias compared to our baseline, lowering the score below zero. Analyzing these conditions together could obscure key differences in the distributions due to the effects evening out.

4.4.1 Analysis of H1, H2 and H3

We created a mixed-effect model to analyze the data. The fixed effects included:

- the strength of initial belief, as calculated from Equation 4.3 (a continuous integer from 0 to 100),
- the data topic (COVID-19 vaccination or diet choices)
- the data facts alignments (*Supporting*, *Refuting*, or the reference level: *No Data Facts*).

Participant ID is the only random effect. The dependent variable is the amount of confirmation bias, as calculated from Equation 4.4. The results are summarized in Table 4.3.

Table 4.3: Mixed-effect linear modeling results. The fourth column, ‘Coefficient P Value,’ represents the p-value obtained from the model fitting process for each category under each independent variable. The fifth column, ‘Variable P Value,’ represents the p-value derived from the Type III ANOVA for each independent variable.

Independent Variable	Coefficient	Std. Error	t-value	Coefficient p-Value	Variable p-Value
Initial Belief Strength	0.473	0.034	13.771	< 0.001 ***	< 0.001 ***
Topic	2.414	1.867	1.293	0.196	0.196
Alignment: Refuting	-6.570	2.600	-2.527	0.012 *	Alignment: < 0.001 ***
Alignment: Supporting	7.417	2.600	2.853	0.004 **	

H1: Prior Beliefs

As shown in Figure 4.5, under **H1** *we found that participants’ confirmation bias score is significantly positively correlated with the strength of their pre-existing beliefs, regardless of the direction of data facts.* With one unit increase in the strength of the pre-existing belief, the confirmation bias score increases by about 0.473 units. This **supports H1**.

H2: Topic

While we observe a trend that the COVID-19 vaccination topic resulted in higher confirmation bias (*Estimated Marginal Mean(EMM) = 20.1*) compared to the diet choice topic (*EMM = 17.7*), we observe no significant effect of topic. Thus **H2 is not supported**. One possible explanation for the lack of significance could be

that the topical differences are captured by prior beliefs (**H1**). In particular, people hold stronger prior beliefs towards polarizing topics ($p < 0.001$, 95% CI for the mean difference in initial belief strength between diet and vaccination topics: [-18.27, -13.30]). As a result, when both prior belief strength and topic are included in the model, the topic itself might not significantly impact confirmation bias.

H3: Data Facts Alignment

Compared to the *No Data Facts* condition, participants exhibited significantly more confirmation bias when reading *Supporting Data Facts* ($EMM_{sup} - EMM_n = 7.4$), while they exhibited significantly less confirmation bias when reading *Refuting Data Facts* ($EMM_n - EMM_{ref} = 6.6$). These results **support H3**.

4.4.2 Analysis of H4

We conducted the statistical analysis under both situations **SN** and **RN**, as outlined at the beginning of Section 4.4. Since this is a between-subject comparison, we constructed a linear model with the following predictors:

- the strength of initial belief (a continuous integer from 0 to 100, calculated using Equation 2),
- the topic of the dataset (COVID-19 vaccination or diet choices)
- the data facts presentation style (textual data facts only, textual data facts with visual annotation, or *No Data Facts*).

The dependent variable was the confirmation bias score. The results are summarized in Table 4.4.

We found a significant effect of display modes on confirmation bias under both **SN** and **RN**. For the situations where participants read the *supporting* data facts **SN**, post-hoc analysis with Bonferroni correction [133] suggests that the significant effect

is driven by the significant difference in confirmation bias levels between the baseline condition (*No Data Facts*) and [Textual Data Facts with Visual Annotations] condition ($p_{\text{SN}} = 0.014$).

For the situations where participants read the *refuting* data facts **RN**, post-hoc analysis suggests only a trending significant difference in confirmation bias level between the baseline and the [Textual Data Facts with Visual Annotations] condition ($p_{\text{RN}} = 0.051$).

We observed no significant difference between the baseline and the [Textual Data Facts Only] condition under both **SN** and **RN**. These results **partially support H4**.

4.4.3 Analysis of H5

We again consider both the situations **SN** and **RN**. We constructed a linear model with the following fixed effects:

- the strength of initial belief (a continuous integer from 0 to 100, calculated using Equation 2)
- the topic of the dataset (COVID-19 vaccination or diet choices)
- the strength of data facts ([Correlation] Data Facts, [Derived Value] Data Facts, or *No Data Facts*)

The dependent variable was the confirmation bias score. The results are summarized in Table 4.4.

We found a significant effect of the data fact strength on confirmation bias under both **SN** and **RN**. Post-hoc analysis with Bonferroni correction suggests that for both *Supporting* and *Refuting* data facts, participants who read the [Correlation] data fact exhibited a significantly **higher** amount of confirmation bias compared to the baseline ($p_{\text{SN}} = 0.004$, $p_{\text{RN}} = 0.002$).

On the other hand, for both *Supporting* and *Refuting* data facts, participants who read the [Derived Value] data fact did not exhibit a significantly different amount of confirmation bias compared to the baseline. These results **partially support H5**.

Table 4.4: Fixed-effects linear modeling results. The sixth column, 'Coefficient P Value,' represents the p-value obtained from the model fitting process for each category under each independent variable. The seventh column, 'Variable P Value,' represents the p-value derived from the Type III ANOVA for each independent variable.

Situation	Independent Variable	Coefficient	Std. Error	t-value	Coefficient p	Variable p
SN	Style: Textual Only	6.348	3.011	2.108	0.035 *	Style: 0.008 **
	Style: Textual + Visual	8.451	2.980	2.836	0.005 **	
	Strength: Correlation	9.688	3.039	3.187	0.001 **	Strength: 0.005 **
	Strength: Derived Value	5.333	2.952	1.807	0.071	
RN	Style: Textual Only	-5.905	3.061	-1.929	0.054	Style: 0.0279 *
	Style: Textual + Visual	-7.229	3.029	-2.387	0.017 *	
	Strength: Correlation	-10.134	2.997	-3.382	0.001 **	Strength: 0.003 **
	Strength: Derived Value	-2.694	3.086	-0.873	0.383	

4.5 Discussion

4.5.1 The Role of Visual Annotation

In our findings on **H4**, we were surprised to observe that *supporting* data facts might only increase confirmation bias when accompanied by visual annotations, suggesting that supporting data facts alone may not significantly bias users. This unexpected finding motivates us to reconsider the role of *visual annotations* in visualization design.

The use of annotations and embellishments in visualizations has long been debated within the visualization community. While some researchers advocate for maximizing the data-ink ratio to reduce chart junk that could distract or mislead viewers [117, 116], others argue that annotations can improve the accuracy of information recalled from visualizations [5, 55]. More recent research demonstrated that annotations can significantly influence people to draw conclusions aligned with the

annotations, beyond what might naturally capture their attention [6, 111]. These insights together provide a potential explanation for our findings on **H4**: namely, visual annotations could enhance the salience of patterns described by textual data facts and thus shape how individuals interpret a visualization as *supporting* or *refuting* evidence, which ultimately affects the manifestation of confirmation bias. Therefore, designers should be aware that when visual annotations are combined with textual summaries, there is a strong potential to consolidate viewers' prior beliefs when there is alignment, making individuals more susceptible to confirmation bias.

4.5.2 (De-)Biased or Echoing?

The opposite effects of *supporting* and *refuting* data on confirmation bias prompt us to consider potential driving factors. Were people prompted to think more critically when they read *refuting* data facts, or did they just echo back the trends described by the data facts? Our results demonstrated that *refuting* data facts cause people's beliefs to shift 35% (See Figure 4.5 H3 Bottom)², resulting in an estimated marginal mean of confirmation bias scores of 12.0, significantly greater ($p = 0.035$, 95% CI [8.59, 15.5]) than a baseline of zero (which means no confirmation bias). If people were simply echoing the perspective of the *refuting* data facts, we would expect greater belief movement in the opposite direction, with sub-zero values for the confirmation bias score. But this was not the case. In other words, people still exhibited confirmation bias (albeit less) even when viewing *refuting* data facts. This finding supports the case that *refuting* data facts help mitigate confirmation bias, rather than the case where participants just echo the trends described by the data facts. This observation also corroborates that confirmation bias is a weighting bias rather than an interpretation bias [73], where people assign higher weight to evidence that *supports* their prior beliefs than to evidence that *refutes* them.

²35% is calculated from the estimated marginal means in Figure 4.5 H3 Bottom as $(18.6-12)/18.6$.

4.5.3 Exacerbating or Mitigating Confirmation Bias?

Consider a viewer who holds a strongly positive belief about COVID-19 vaccination. If they encounter a data visualization with data facts that *support* their existing belief, it could strengthen their conviction by 40% (See Figure 4.5 H3 Top) However, if they are presented with data facts that *refute* their prior belief, it could weaken their conviction, shifting their belief by 35% in the opposite direction (See Figure 4.5 H3 Bottom)³. This suggests that *refuting* data facts might help reduce confirmation bias. While the distribution of bias scores with *refuting* data facts remains positive (suggesting that some confirmation bias is still present), it is nevertheless lower than in scenarios with *No Data Facts*, indicating a potential trend toward reducing confirmation bias with *refuting* data facts relative to the baseline level of bias. Our findings corroborate previous research in data visualization that highlights the power of text alongside data visualization to shape individuals’ interpretations of data [111, 59, 65]. These findings might also indicate that providing alternative interpretations of the same visualization—especially those that challenge prior beliefs—might offer a more balanced perspective of the data and might prevent decisions based solely on *supporting* evidence.

Data Facts Consideration Our experimental findings revealed that data facts have potential both to exacerbate or mitigate confirmation bias depending on whether they are framed as *supporting* or *refuting* the viewer’s prior beliefs. Realistically, as the use of large language models proliferates through more application areas, it will be hard to avoid the inevitable combinations of visualizations supported by automatically (or manually) generated textual summaries. Rather, we assert that this is an opportunity to leverage the mitigating potential of data facts in future systems. In particular, if future systems leverage belief elicitation techniques [68] prior to showing

³40% is calculated from the estimated marginal means in Figure 4.5 H3 Top as $(26-18.6)/18.6$, and 35% from the means in Figure 4.5 H3 Bottom as $(18.6-12)/18.6$.

users data, systems can intelligently display refuting data facts alongside visualizations, thereby mitigating confirmation bias in interactive settings. Nevertheless, there is a clear risk that malicious actors could misuse these insights to induce a favorable bias. We encourage responsible use of these insights, not as a tool for unfair manipulation, but as a tool to promote balanced thinking and informed decision making.

4.5.4 Refuting Data Facts and Counterfactuals in Visualizations

Existing work in visualizations have examined other approaches to reduce confirmation bias in visual data analysis. One such approach involves the use of counterfactuals, which refer to the mental constructs that represent *alternatives* to reality [17, 97]. When people think "if only" or "what if," they imagine how the past could have unfolded differently. This process helps reduce bias by encouraging individuals to consider alternative factors that could have led to the same outcome. In an analytical scenario where the data suggests two variables might be causally related, counterfactuals encourage users to question whether the two factors are truly causally related or if a third variable might be influencing the results [127, 128, 129]. In another example, highlighting points that have previously been examined (thereby visually distinguishing points that have/have not been previously examined) when analyzing data with visualizations encourages users to focus more on unexplored areas to potentially gain new insights [126, 42]. Further, providing people with *refuting* data facts, as we did in our study, can also prompt people to consider alternative perspectives. Thus, counterfactuals, highlighted data values, and *refuting* data facts offer users alternative perspectives that challenge their existing beliefs about the data. These insights underscore the potential de-biasing power of incorporating multiple data perspectives within a visualization. By empowering analysts to consider a wider range of alternatives for a balanced analysis, these techniques can combat over-reliance on

existing beliefs and reduce confirmation bias.

4.5.5 Limitations

One limitation of our study is the narrow set of topics tested. We selected only one polarizing topic and one neutral topic, with their levels of polarization evaluated by a small group of 40 participants. However, there are other topics where people overwhelmingly share the same stance, such as the general affection people have for pets [4]. Future research should test a wider range of topics to further understand the generalizability of our findings.

Another limitation is the use of synthetic datasets. While these datasets allowed us maximum control over our experiment, they may not fully capture real-world scenarios relevant to the topics studied. For example, a participant who is particularly savvy in the topic may suspect the authenticity of the dataset and evaluate associated evidence in unpredictable ways. Additionally, in the present study, we operationalized confirmation bias as interpreting ambiguous information in a way that aligns with one’s prior belief, and thus, designed visualizations to be intentionally ambiguous. In the real world, not all visualizations would be designed to be ambiguous; some are explicitly crafted to highlight strong data patterns and convey a particular narrative. Confirmation bias would undoubtedly happen with these visualizations as well. Existing work by Xiong et al [139] have demonstrated that in these cases, visual representation formats such as showing the data via a table or a bar, would change the strength of confirmation bias. Therefore, we encourage future work to also investigate the effect of data facts and visual annotations on confirmation bias in these scenarios where the visualization stimuli are not ambiguous.

Furthermore, the real-world factors related to the statements in Table 4.2 are likely more complex than what is depicted in the data. For instance, in the COVID-19 Vaccination topic, participants might also consider factors such as mortality rates across

different age groups to determine if the vaccination should be mandatory, while in the Diet Choices topic, other health metrics beyond the risk of chronic disease—such as Body Mass Index (BMI), nutrition levels, and emotional well-being—are also indicators of diet health when making comparisons. The limited factors provided in our data might introduce distortions in users’ evaluations that are, to some extent, inevitable in controlled studies. Future work can incorporate more realistic datasets to generalize findings in ecologically valid settings that better reflect the complexity of real-world decision-making processes.

Finally, we also determined the strength of the data facts solely based on modulating which visual patterns they described (*PS 3*, Section 4.2.3). Further research could explore how factors such as the phrasing of the data facts, source credibility, and trustworthiness influence their perceived strength.

4.6 Summary

In this chapter, we explored the impact of presenting textual data facts alongside visualizations and uncovered key insights about confirmation bias. Participants with stronger-held prior beliefs were more likely to interpret ambiguous visualizations as supporting their prior beliefs. When visualizations were accompanied by textual data facts, confirmation bias was *exacerbated* (when visualizations were paired with *supporting* data facts) and *mitigated* (when paired with *refuting* data facts). Confirmation bias was further exacerbated when data facts focused on *correlations* compared to *average values*. Our results also suggest that visually annotating *supporting* data facts further amplified these effects. These findings together indicate that both the design of the accompanying data facts and an individual’s prior beliefs play crucial roles in shaping how visual information is interpreted. Our research highlights the complex interplay between visual and textual evidence in shaping data interpretation.

Particularly with recent advances in AI-generated content, we hope these results will stimulate further discussion and serve as a catalyst for a research agenda within the community to assess the potential benefits and drawbacks of combining visualizations with data facts.

Chapter 5

In Progress: Confirmation Bias through Outlier Identification

In this chapter, we present our ongoing work in response to **Research Question 3**, which extends the characterization of confirmation bias from the static visualization setting as in Research Question 2 to an **interactive** visualization setting.

RQ 3 *How does confirmation bias manifest through user interactions with visualizations?*

In visualization systems, interaction is just as essential as visual representations – both are fundamental components that facilitate a dialog between the system and its users, enabling insight discovery from data [142]. At the same time, interaction serves as an externalization of the user’s thought process, supporting their reasoning by helping them break down and make sense of complex data efficiently [124]. Since interactions in visualization systems might actively modify visual representations [142], it also opens the door for biased behaviors [124, 125, 126]. When users hold strong prior beliefs about a data topic, they may fall into confirmation bias [75, 15], using interaction to manipulate the view—such as by only selecting or hovering over data points that support their beliefs, or filtering out those that contradict

them. In doing so, they shape the visual output to align with their prior beliefs, thereby reinforcing biased interpretations and undermining objective data analysis.

The goal of this work is to investigate to what extent users exhibit confirmation bias through their filtering interactions with visualization when they hold strong prior beliefs about the data. We aim to summarize our findings as a full paper to ACM CHI 2026.

5.1 User Study

5.1.1 Motivation

One manifestation of confirmation bias is the selective search for information that supports one’s prior beliefs [88]. For example, individuals may seek out news articles or online search results that align with their beliefs while ignoring those that present opposing views. This tendency can also influence how users interact with data visualization. For example, through data exploration, people may filter out data that does not agree with their pre-existing hypotheses. Specifically, we hypothesize that:

Confirmation bias could manifest in how individuals perceive the “expected relationship or pattern” of the dataset, especially when they hold strong prior beliefs on the dataset topic. This bias could, in turn, influence their judgment of anomalies, making them more likely to identify deviations from belief-driven expectations rather than from the statistical distribution of the data.

To validate our hypotheses, we adapted the “Find Anomalies” task from the framework by Amar et al., asking participants to visually identify outliers in a 2D scatterplot. Specifically, we focused on participants’ use of the **Select** interaction – one of the interaction types defined in Yi et al.’s visualization taxonomy [142] – which

involves marking data points of interest for subsequent analysis. Our goal was to examine whether participants tend to **Select** data points that deviate from their expected linear relationship, particularly in cases where the underlying relationship in the scatterplot is ambiguous or weak. Beyond capturing a confirmation bias scenario in the context of interactive visualization, our study also offers practical insights for data science practitioners. In particular, it highlights how confirmation bias can influence data cleaning – a crucial step where outliers are often identified through visual inspection and may be removed or replaced. When outlier handling is guided by subjective expectations rather than objective criteria, it can introduce bias into the dataset and lead to flawed conclusions in subsequent analyses. This underscores the need to recognize and address confirmation bias early in the data analysis process.

5.1.2 Belief Elicitation Methodology

Establishing the user’s prior beliefs is a critical step towards understanding how confirmation bias manifests – e.g., do individuals then interpret ambiguous evidence in a way that aligns with those beliefs? To establish prior beliefs, we need a method to elicit those beliefs that will allow users to map their internal beliefs to a fixed scale that will allow us to conduct quantitative analyses. To capture how users internally represent their beliefs in relation to the stimulus – a 2D scatterplot – we adopt a visual belief elicitation technique inspired by Koonchanok et al. and Karduni et al., which allows participants to express both the direction and uncertainty of their prior belief as a linear relationship in two steps [58, 68].

- Step 1: Participants first interact with a line whose slope can be adjusted between -1 and 1 via a slider to indicate the perceived direction of the relationship between vaccination rate and mortality rate (μ).
- Step 2: While the line in step 1 remains visible, participants are then asked to

indicate their uncertainty about this belief using a separate slider ranging from 0 to 1 (σ).

The values from both sliders – the direction (μ) and the uncertainty (σ) – are subsequently used to parameterize the following linear model, which we used to animate a sequence of hypothetical outcome plots (HOPs) to help participants reflect on their judgment of both step 1 and step 2. We modeled the relationship between the x and y coordinates as a linear function, as shown in Equation 5.1.

$$y = \beta x + \beta_0 \quad (5.1)$$

In Equation 5.2, β is a noisy slope of the linear relationship between x and y , modeled as a normally distributed variable. The mean of the distribution (μ) corresponds to the slope specified by each participants in Step 1 as the direction of the prior belief, while the standard deviation is given by σ , which each participant provided in Step 2 to indicate the uncertainty of the prior belief. This relationship is further illustrated in Equation 5.2.

$$\beta \sim \mathcal{N}(\mu, \sigma^2) \quad (5.2)$$

The inclusion of an additional residual β_0 is to prevent the generation of a perfectly fitted line even when participants report high confidence (i.e., $\sigma = 0$). As depicted in Equation 5.3, β_0 is also a normally distributed variable with a fixed standard deviation of $\sigma_b = 0.1$. Following the approach of Koonchanok et al. [68], we use this model to animate a sequence of hypothetical outcome plots (HOPs) – updated every 200 milliseconds – to help participants perceive the range of plausible data patterns implied by their indicated relationship directions and associated uncertainties. A depiction of three animation frames illustrating different levels of uncertainty in the

expected relationship is shown in Figure 5.2.

$$\beta_0 \sim \mathcal{N}(0, \sigma_b^2) \quad (5.3)$$

5.1.3 Study Procedure

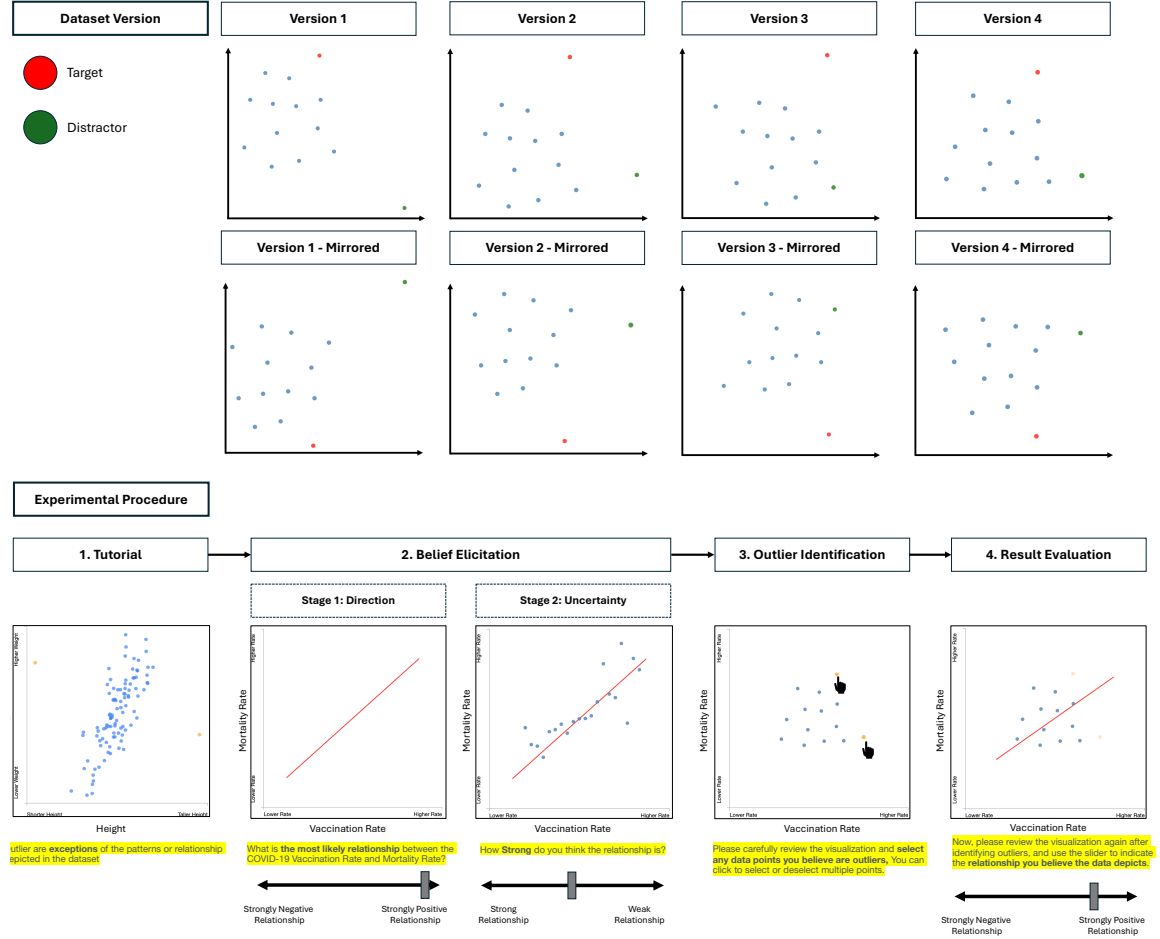


Figure 5.1: Experiment Conditions and Procedure: [Dataset Version] is the only between-subjects condition. In Step 3 of the [Experimental Procedure]: Outlier Identification, each participant will be randomly assigned to one of eight versions of the dataset.

A summary of the study procedure and the stimulus design is shown in Figure 5.1. After completing the consent form, participants will be guided through a tutorial that introduces the concept of outliers and familiarizes them with the visual interface used

for outlier identification. The tutorial begins with a scatterplot containing 100 data points that depict a general positive correlation between high school students' height and weight, along with two outliers: one student with unusually short height and high weight, and another with unusually tall height and low weight. To avoid biasing the way participants identify outliers, these points are introduced as examples of outliers – defined as “exceptions to the relationship” between height and weight, following the general definition of anomalies proposed by Amar et al. [1]. After this introduction, participants will be asked to identify the two outliers by clicking and highlighting them to complete the tutorial.

Following the tutorial, participants will be asked to express their prior belief about the relationship between vaccination rate and mortality rate, following the setup described in subsection 5.1.2. After the belief elicitation stage, participants will proceed to identify outliers in an interactive scatterplot. Participants will be able to select and highlight potential outliers by clicking on individual data points and deselect them by clicking again. They will be free to select as many or as few outliers as they deem appropriate.

As the final step of the main study, participants will evaluate the resulting visualization by specifying the perceived slope of the scatterplot (μ_e) and answering a multiple-choice question designed to capture their rationale for selecting those outliers. To ensure data quality, two checks will be included: a comprehension check immediately after the tutorial, which asks participants to identify the correct definition of outliers through a multiple-choice question; and an attention check following the evaluation stage, which states, “*What is your favorite color? This is an attention check, please choose purple if you are paying attention.*” Participants who fail either the comprehension check or attention check will be excluded from the data analysis.

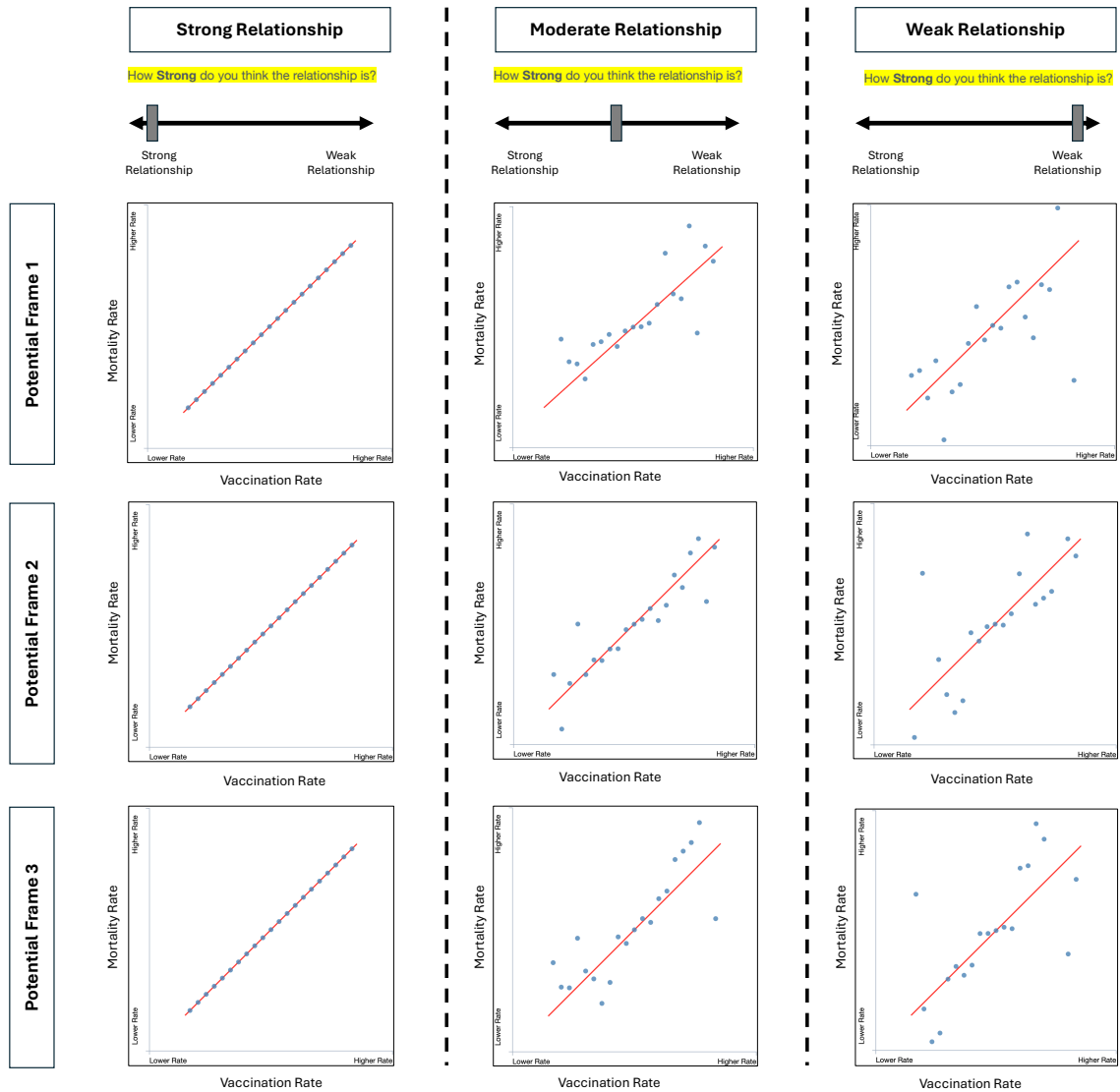


Figure 5.2: An illustration of potential frames from the HOPs animation under varying strengths of relationships.

5.1.4 Stimulus Design

As hypothesized in subsection 5.1.1, we expect participants to be more likely to identify outliers when doing so produces a data pattern that aligns with their prior beliefs. To capture this behavior, we aimed to design visualization stimuli as a 2D scatterplot that deliberately introduced ambiguity – such that removing different outlier data points could reveal either belief-consistent or belief-inconsistent linear relationships. This design allows us to directly observe confirmation bias, as participants selectively identify outlier data points whose exclusion results in a linear relationship between x and y that aligns with their prior beliefs.

To guide our stimulus design, we established three key principles. First, the visualization must be deliberately ambiguous and avoid presenting an obvious linear relationship; otherwise, participants are likely to converge on similar interpretations, making it difficult to detect confirmation bias.

Principle 1: The dataset should be ambiguous enough to prevent a clearly dominant linear relationship, allowing for interpretations aligned with different prior beliefs.

Secondly, we design the dataset to include “underlying truth” outlier data points – statistically defined outliers that participants are expected to identify if they focus solely on the presented data patterns, independent of their prior beliefs. We define outliers as data points with studentized residuals greater than 2, following the recommendation by Blatna et al. for data points in a two-dimensional dataset [7]. This setup allows us to distinguish between belief-driven and data-driven behavior: participants influenced by their prior beliefs may overlook true outliers if those outliers contribute to a belief-consistent pattern, and instead focus on removing data points whose exclusion reinforces their prior beliefs. In contrast, participants who rely more on the data itself are more likely to identify the true outliers—even if doing

so produces a visual pattern that conflicts with their existing beliefs.

Principle 2: The dataset should include legitimate outlier(s) to ensure participants can still identify valid anomalies based solely on the underlying data pattern.

Finally, to further prevent uniform responses across participants, we also need to carefully consider the spatial arrangement of outliers. If the outliers are too visually salient – particularly after participants undergo training – they may systematically remove all of them, thereby reducing the intended ambiguity of the task.

Principle 3: Outliers should not be overly obvious.

After establishing these principles, we recognized a tension between Principle 1 and Principle 3: making outliers less visually obvious can reduce their influence on the overall data pattern – such that the patterns within the visualization appears largely unchanged regardless of whether the outliers are present. To address this challenge, we introduce an additional layer of ambiguity by incorporating fake outliers (distractors) – data points that are not statistically defined as outliers but appear visually distant from the main data cluster. We hypothesize that participants are more likely to remove these fake outliers when the remaining data pattern aligns with their prior beliefs, and more likely to examine the pattern of the whole dataset and identify the real outlier (target) when it contradicts them. To evaluate this hypothesis, we created four variations of such datasets by manually assigning coordinates to 14 data points – comprising 1 target, 1 distractor, and 12 additional normal points – while ensuring all design principles were satisfied. We began with a small number of data points in our pilot study to (i) make the visualization easier for participants to interpret, and (ii) allow us to focus on the underlying dataset structure to assess whether any of the four versions could elicit diverse outlier identification behaviors. We additionally implemented a mirrored version of each dataset by flipping the y-axis of all data

points to ensure full coverage in our testing. The stimuli for all versions, including the mirrored ones, are shown in the upper part of Figure 5.1.

Version 1: Both the real and fake outliers are visually obvious.

Version 2: Only the fake outlier is visually obvious.

Version 3: Only the real outlier is visually obvious.

Version 4: Neither the real nor the fake outlier is visually obvious.

We plan to conduct a pilot analysis to identify which version elicits the greatest variation in interpretations, and use that version in our formal study.

5.2 Hypotheses and Analysis Plan

To validate the main hypotheses proposed in subsection 5.1.1, we decompose them into three specific sub-hypotheses, each aligned with an expected outcome of our study design:

- H1.** Participants will remove more outliers when the overall dataset initially contradicts their prior beliefs.
- H2.** When the fake outlier and the normal data points together support a participant's prior belief, the participant is more likely to identify and remove the real outlier; conversely, when the real outlier and normal data points align with the prior belief, they are more likely to remove the fake outlier.
- H3.** Participants who express lower uncertainty during the belief elicitation stage will produce final slope estimates that are more closely aligned with their prior beliefs – reflected in a smaller absolute difference between the elicited slope (μ) and the evaluated slope after filtering (μ_e).

To test these hypotheses, we will conduct the following analyses:

- **H1: Outlier Quantity and Belief Alignment.** We will compute the number of outliers removed by each participant. A linear regression model will be used to predict the number of outliers removed. The key independent variable will be the alignment between the participant’s prior belief (μ) and the slope of the original dataset (μ_d). A greater misalignment (e.g., opposite directions) is expected to predict more outlier removals. A significant positive coefficient will indicate the support of H1.
- **H2: Outlier Type and Belief-Consistent Interpretation.** We will use two separate logistic regression models to predict whether the real outlier and the fake outlier were removed (dependent variable: `removed_real` or `removed_fake`, coded as 1 if the outlier was removed and 0 otherwise). The key independent variable will be the change in slope alignment: the absolute difference between the participant’s prior belief slope (μ) and the slope of the dataset after removing the outlier (μ_{post}), minus the absolute difference between the prior slope and the original dataset slope (μ_d):

A positive value of this variable indicates that removing the outlier brings the dataset’s slope closer to the participant’s prior belief. We hypothesize that participants will be more likely to remove an outlier when this value is larger – i.e., when the slope after removal more closely aligns with their belief. Thus, a significant positive coefficient on `slope_diff_change` would support H2.

$$\text{slope_diff_change} = |\mu - \mu_d| - |\mu - \mu_{post}| \quad (5.4)$$

A significant positive coefficient on `slope_diff_change` will indicate that participants are more likely to remove a real/fake outlier when doing so produces a data pattern that better supports their prior belief – thus supporting H2.

- **H3: Belief Uncertainty and Evaluation Alignment.** To test the relationship between belief uncertainty and belief-consistent evaluation, we will compute the absolute difference between the participant’s prior belief slope (μ) and their evaluated slope after filtering (μ_e). A linear regression model will then predict this difference using the uncertainty value (σ) from the belief elicitation stage.

$$|\mu - \mu_e| \sim \sigma \quad (5.5)$$

A significant negative coefficient on σ (or a positive coefficient on confidence $= 1 - \sigma$) will indicate that lower belief uncertainty is associated with greater belief reinforcement.

We will conduct a power analysis using the observed effect sizes for each hypothesis to determine the required sample size for our formal experiment, aiming for a statistical power of 0.8.

5.3 Summary

In this chapter, we outlined our ongoing work addressing RQ3, presenting the experimental design and quantitative analysis plan for a pilot study that investigates how individuals reinforce their prior beliefs through *interactions* with data visualizations.

Chapter 6

Proposed Work: Mitigating Confirmation Bias Through Visual Encoding Manipulations

This section describes my proposed research in response to

RQ 4 *How can visualization design choices be used to mitigate confirmation bias?*

Research in psychology and visualization has demonstrated that confirmation bias is notoriously difficult to overcome [140]. Even awareness of this bias provides minimal protection [63], largely due to individuals’ deeply held prior beliefs [88]. Recent visualization research suggests that systematically adjusting visual encodings—specifically, adjusting the size and opacity of data points based on their residual distance from a regression line—can significantly alter how users perceive correlations in scatterplots [113]. As depicted in Figure 6.1, by visually emphasizing data points closer to the regression line, this technique increases the perceived strength of correlation, prompting users to update their beliefs about the relationship between the variables.

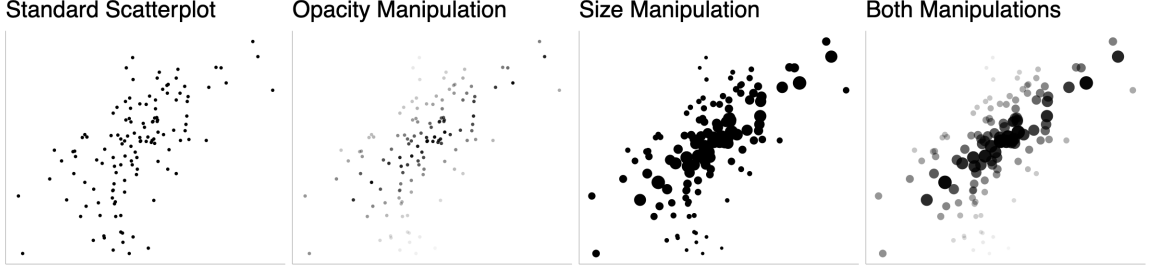


Figure 6.1: The alternative scatterplot design with size and opacity manipulation proposed by Strain et al. [113]

Building on this insight, my proposed work seeks to leverage visual encoding manipulations alongside prior belief elicitation to mitigate confirmation bias. By first eliciting users' prior beliefs about the relationship between variables, visual encodings can then be systematically adjusted to either emphasize or de-emphasize correlations accordingly. Specifically, data points contradicting users' prior expectations could be visually highlighted by increasing their size and opacity, whereas points aligning with their existing beliefs could be visually de-emphasized. We will implement our intervention by adapting the opacity and size manipulation techniques introduced by Strain et al. [113], as detailed in Equation 6.1 and Equation 6.2.

$$point_{\text{size}} = 1 - b^{\text{residual}} \quad (6.1)$$

$$point_{\text{opacity}} = 1 - b^{\text{residual}} \quad (6.2)$$

Specifically, Strain et al., employed the same mechanism to adjust both the opacity and the size of each data point in the scatterplot. As shown in both Equation 6.1 and Equation 6.2, b is a value strictly between 0 and 1, and *residual* denotes the vertical distance between each data point and the regression line fitted to the entire dataset. Therefore, under this mechanism, data points closer to the regression line will appear smaller and more transparent, whereas those farther away from the line will appear larger and more opaque.

Inspired by our findings in chapter 4, which showed that the strength of prior

belief is a strong predictor of confirmation bias, we will design this intervention to correspond proportionally to the strength of prior beliefs. Specifically, for users holding stronger prior beliefs, a larger b value in Equation 6.1 and Equation 6.2 can be used to create a more pronounced visual intervention, providing stronger mitigation of confirmation bias. This targeted approach can gently direct users’ attention toward evidence challenging their prior beliefs, reminding them of their potential biases while maintaining uninterrupted data exploration.

To achieve this goal, I am collaborating with Dr. Cindy Xiong Bearfield, an expert in visual perception, cognition, and data visualization at Georgia Tech. Together, We aim to summarize our findings as a full paper at IEEE VIS 2026.

6.1 User Study

We will conduct user studies to investigate the extent to which tailoring visual encodings in scatterplots – specifically the size and opacity of data points – based on strength of participants’ prior beliefs, can influence the degree of confirmation bias exhibited when estimating the correlation of scatterplots. We will recruit participants from the crowdsourcing platform Prolific [90]. The number of our participants will be determined based on the effect size observed in a pilot study using the design below, targeting a statistical power of 0.8.

6.1.1 Study Procedure

Participants will begin by expressing their prior beliefs about the dataset topic, following the same two-step procedure described in subsection 5.1.2. Specifically, they will use an adjustable line with a slope ranging from -1 to $+1$ to indicate the direction of their belief (μ), and adjust a slider from 0 to 1 to indicate the uncertainty of their belief (σ).

Next, participants will evaluate one of two scatterplot conditions: (a) a standard version without any visual encoding manipulations, or (b) a version in which the size and opacity of each data point is dynamically adjusted according to Equation 6.3 and Equation 6.4. This formula is an adaptation of the visual encoding manipulation used by Strain et al., designed to *reduce* the visual salience of the data points that are more consistent with participants' prior beliefs.

$$point_{\text{size}} = 1 - \left(\frac{\sigma}{2}\right)^{residual} \quad (6.3)$$

$$point_{\text{opacity}} = 1 - \left(\frac{\sigma}{2}\right)^{residual} \quad (6.4)$$

In both Equation 6.3 and Equation 6.4, *residual* denotes the the vertical distance between each data point and the regression line with the slope corresponding to each participant's stated belief direction (μ), and σ reflects the uncertainty of that belief obtained from belief elicitation stage. Because σ represents participants' uncertainty on a continuous scale from 0 to 1 (as established by Koonchanok et al. [67]), we scale σ by dividing it by 2. This ensures that the expected value of participants' reported uncertainty ($\sigma = 0.5$) corresponds to the fixed manipulation strength ($b = 0.25$ in Equation 6.1 and Equation 6.2) used by Strain et al. [112, 113], which was shown to effectively modulate the visual salience of data points. By doing so, our dynamic manipulation remains anchored to an empirically validated setting, while still allowing the strength of visual encoding to vary proportionally with participants' expressed belief strength. To facilitate assessment of confirmation bias – operationalized as the degree to which participants' estimated correlations align with their prior beliefs – all scatterplots are generated from the same underlying dataset with a true correlation of 0.

Finally, participants will estimate the Pearson correlation coefficient (r) of the scatterplot. To ensure they are familiar with the concept, the study will begin with a

brief tutorial on Pearson’s r , followed by interactive practice using a slider to estimate correlations from example scatterplots.

6.2 Hypotheses and Analysis Plan

Throughout the study design, we hypothesize that:

- H1** Participants will exhibit confirmation bias by evaluating the correlation of the scatterplot in a direction that aligns with their prior beliefs, even when the data does not support such a conclusion.
- H2** Participants who express stronger prior beliefs will exhibit a greater level of confirmation bias.
- H3** Participants who reason with a version of the scatterplot where visual encodings (size and opacity) are manipulated to downplay belief-consistent visual patterns will exhibit a lower level of confirmation bias compared to those who view the standard version.

For each participant, we will compute a confirmation bias score that captures both the magnitude and direction of deviation in their estimated correlation ($\hat{r}$) relative to the ground truth correlation of the dataset, which is fixed at $r_d = 0$. The sign of the score indicates whether the estimated correlation aligns with the direction of the participant’s prior belief slope (μ), and the magnitude reflects the strength of confirmation bias.

$$\text{bias_score} = \pm |\hat{r}| \tag{6.5}$$

To validate H1, we will test whether the average `bias_score` is significantly greater than 0 using a one-sample t -test. A positive mean would indicate a tendency to interpret the data in a direction consistent with participants’ prior beliefs, thus supporting H1.

To validate H2 and H3, we will fit a linear mixed-effects model with `bias_score` as the dependent variable, `condition` (baseline vs. manipulated) and `belief_strength` (the σ value) as fixed effects, and a random intercept for each participant to account for repeated measures. If the visual encoding manipulation is effective (H2), we expect to observe a negative and significant effect of the manipulated condition on `bias_score`, indicating reduced confirmation bias. Similarly, if belief strength influences confirmation bias level (H3), we anticipate a positive and significant effect of `belief_strength`, suggesting that stronger prior beliefs are associated with greater bias.

6.3 Summary

In this chapter, we proposed work to address RQ4 by outlining a visual encoding manipulation mechanism – motivated by the work of Strain et al. [112, 113] – designed to mitigate confirmation bias during the interpretation of scatterplots. We also described an experimental design and quantitative analysis plan to assess the effectiveness of these interventions.

Chapter 7

Discussion

7.1 The Impact of Belief Elicitation on Confirmation Bias

The goal of chapter 3 is to understand how individuals represent their internal beliefs through visualization. These insights can inform the design of belief elicitation techniques that strike a balance between the analyzability and expressive richness of elicited data. However, the elicitation process itself – particularly in a visual form – may inadvertently reinforce prior beliefs [68]. This introduces a compelling tension: to evaluate confirmation bias, we must first elicit individuals’ prior beliefs, yet the influence of the elicitation process on confirmation bias being measured remains insufficiently understood.

Prior research in visualization has demonstrated that eliciting individuals’ prior beliefs before presenting data can promote more effective analysis and lead to more accurate interpretations [67, 68]. As noted by Koonchanok et al., this process may activate participants’ prior knowledge, making them more critical of implausible data patterns. Moreover, explicitly articulating one’s beliefs can engage cognitive mechanisms akin to self-explanation [60], which can deepen understanding, especially when

individuals are encouraged to map the strength of their beliefs to data scenarios that plausibly align with those beliefs. At the same time, Koonchanok et al. also caution that belief elicitation could inadvertently heighten confirmation bias: participants whose priors were elicited were more skeptical of data samples that contradicted their beliefs than those whose priors were not elicited. Therefore, we encourage future researchers to further investigate to what extent the process of belief elicitation influences confirmation bias. Specifically, future research could conduct a between-subjects study in which participants express their prior beliefs either *before or after* interpreting a visualization. In both conditions, the elicited prior beliefs can be used to quantify the magnitude of confirmation bias, enabling a comparison of whether eliciting beliefs prior to data exposure influences the extent of exhibited confirmation bias.

7.2 Other Forms of Confirmation Bias

Across chapter 4, chapter 5, and chapter 6, we focus on identifying and mitigating confirmation bias during the *evidence interpretation stage*, examining how individuals make sense of visualizations that can support multiple interpretations. Real-world data are often noisy and ambiguous, and because people typically have limited access to the underlying mechanisms that generate the data, they tend to draw on their prior experiences to interpret the evidence [63].

However, confirmation bias can also emerge earlier – during the *evidence searching stage*[63]. When evaluating hypotheses, individuals often gather information in ways that prioritize minimizing cognitive effort and maintaining their self-schema, rather than uncovering the objective truth[3, 2, 63]. As a result, they are more likely to consider supportive facts over refuting ones during the evidence-searching process.

Confirmation bias continues to play a role during the *information recall stage*,

where individuals tend to remember information that aligns with their existing beliefs [44]. As noted by Vedejová et al., this pattern can be understood through schema theory, which suggests that self-schemas guide not only the interpretation of new information but also the selective recall of schema-consistent content [122, 45]. These recalled memories may then re-emerge during subsequent evidence searches, perpetuating the cycle of confirmation bias.

As summarized in Figure 7.1, future visualization research could also explore identifying confirmation bias in the stage of data search and data recall. For instance, future work could examine how individuals search for evidence within a visualization – particularly as modern visualization systems increasingly incorporate Large Language Models (LLMs) to generate instant insights aimed at enhancing the efficiency of data exploration [130, 134]. An open question is whether users tend to selectively focus on insights that align with their prior beliefs, potentially reinforcing confirmation bias. In parallel, prior work on visualization memorability highlights the influence of design choices such as text, pictograms, and redundant encodings in making visualizations more recognizable and easier to recall [8, 9]. Building on this, future studies could also show participants visualizations containing both belief-consistent and belief-inconsistent patterns, vary design choices like titles, pictograms, or redundant encodings, and then assess which type of information is recalled more accurately and frequently – and whether each design choice enhances recall for one side over the other.

7.3 Confirmation Bias and Metacognition

Research in psychology and visualization has shown that confirmation bias is hard to overcome [140]. To address this challenge, we proposed highlighting belief-contradicting textual summaries in chapter 4 to encourage attention toward opposing data patterns,

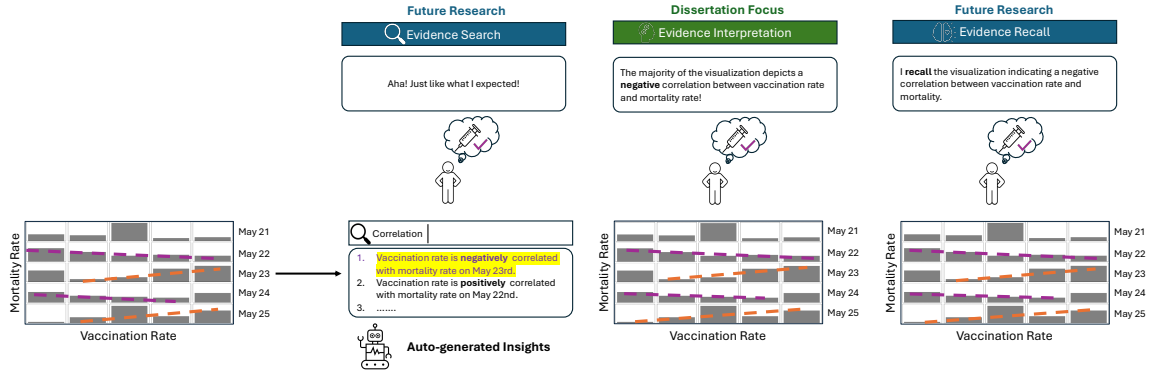


Figure 7.1: Manifestation of Confirmation Bias Across Evidence Search, Interpretation, and Recall Stages in Visualization.

and introduced visual encoding manipulations in chapter 6 to enhance the salience of such patterns. However, cognitive biases – including confirmation bias – are not always bad; they often stem from heuristic decision-making processes that enhance efficiency and reduce cognitive load in complex environments [123]. This perspective invites a more nuanced view: might these interventions (e.g., use of textual data facts, visual encoding manipulation) be unnecessary – or even counterproductive – for individuals who are already skilled in monitoring and reflecting on their reasoning processes?

Through a simulation study in 2021, Rollwage et al. argued that selective information processing – a form of confirmation bias – can be beneficial for simulated agents with strong metacognitive abilities, defined as the capacity to monitor and evaluate the accuracy of one’s own judgments [36]. Specifically, Rollwage et al. [98] implemented two types of agents tasked with inferring the mean (μ) of a normal distribution by choosing between two options: -1 or 1 , based on a sequence of samples drawn from the ground truth distribution. Each agent made an initial decision based on the sign of the first sample. The first was a Bayesian agent, which treated each data point – regardless of whether it supported or contradicted the initial decision – with equal weight. The second was a metacognitive agent, which computed an internal confidence estimate based on the probability of being correct. Crucially, this agent

used its confidence to regulate the degree of confirmation bias: when confidence was high, it selectively downweighted belief-contradicting evidence, reinforcing its initial judgment against noise; when confidence was low, it integrated disconfirming evidence more fully, allowing for further correction. In a simulation of 200,000 trials, agents with high metacognitive efficiency significantly outperformed the unbiased Bayesian agent in correctly inferring the true mean (μ) of the underlying distribution. This work suggests that confirmation bias, when regulated by high metacognitive ability, can serve as a powerful mechanism for filtering out noise in new information throughout the decision-making process.

These findings also offer a promising direction for visualization research. Recent work has begun to explore how enhancing users’ metacognitive awareness during visual reasoning may help mitigate cognitive biases [22]. Building on this direction, future work could also examine whether interventions that adaptively respond to users’ metacognitive abilities – such as their confidence or self-assessed uncertainty – can provide more personalized and effective support for addressing confirmation bias.

Chapter 8

Timeline

The current progress of this dissertation research is summarized in Table 8.1.

Research Question	Status	Publications
1 Representation of Beliefs	✓	[77] (In Preparation)
2 Role of Text and Annotations	✓	[75]
3 Confirmation Bias through Interactions	⌚	[76] (In Preparation)
4 Bias Mitigation through Encoding Manipulation	⌚	

Table 8.1: A summary of the status of this dissertation research. Work that has been completed is indicated by ✓ and work that is in progress is indicated by ⌚.

The anticipated timeline for completing the remaining components of the dissertation is as follows:

- **Fall 2025:** Complete the user study on outlier detection/removal to understand confirmation bias through interactions. We aim to submit our findings to ACM CHI 2026.
- **Spring 2026:** Conduct the user study on mitigating confirmation bias through visual encoding manipulation. We aim to submit our findings to IEEE VIS 2026.
- **Summer 2026:** Complete the dissertation write-up and defense.

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